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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

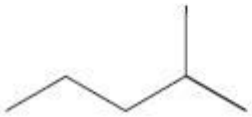
A compound with the formula C_6H_{14} gives a ^{13}C NMR spectrum with four peaks.

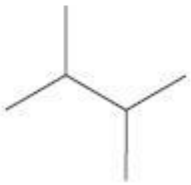
Which compound gives this spectrum?

☐ **A** 

☐ **B** 

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☐ **C** 

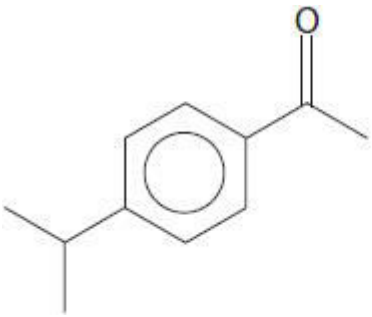
☐ **D** 

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Cuminone is used as a flavouring.



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cuminone

What is the molecular formula of cuminone?

☐ **A** $C_{10}H_{13}O$
☐ **B** $C_{10}H_{15}O$
☐ **C** $C_{11}H_{14}O$
☐ **D** $C_{11}H_{16}O$

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

Q3.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is a redox reaction?

☐ **A** $Cr_2O_7^{2-} + 2C \rightarrow Cr_2O_3 + CO_3^{2-} + CO$

☐ **B** $CrO_2Cl_2 + 4OH^- \rightarrow CrO_4^{2-} + 2Cl^- + 2H_2O$

☐ **C** $Cr_2O_7^{2-} + 2HCl \rightarrow 2CrO_3Cl^- + H_2O$

☐ **D** $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which species contains an element with the same oxidation number as bromine in $KBrO_3$?

☐ **A** N_2O_5

☐ **B** $KMnO_4$

☐ **C** K_2FeO_4

☐ **D** Na_2SO_3

(Total for question = 1 mark)

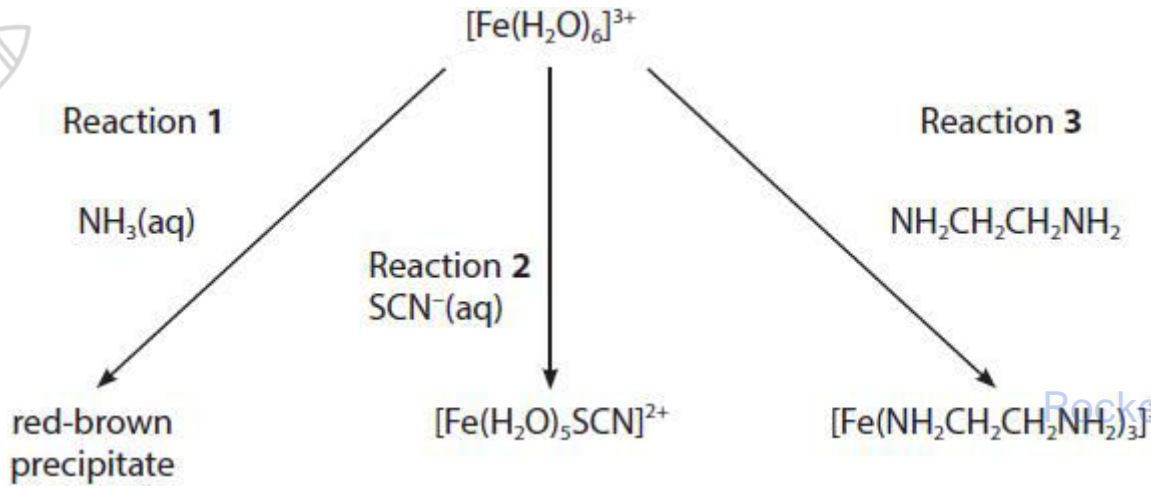
Q5.

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This question is about transition metals and their compounds.

(a) Three reactions were carried out on a solution containing $[Fe(H_2O)_6]^{3+}$ ions.



(i) Complete the equation for Reaction **1**. State symbols are not required.

(2)

$[Fe(H_2O)_6]^{3+}$ + ... NH_3 →

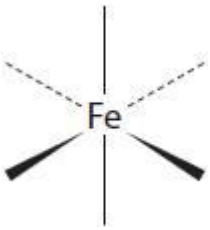
(ii) State the type of reaction occurring in Reaction **2**.

(1)

.....

(iii) Complete the diagram to show the structure of the complex ion formed in Reaction **3**, showing **all** of the atoms.

(2)



(b) Vanadium can exist in the oxidation states +2, +3, +4 and +5.

(i) Complete the table to show the colours of the ions that each of these oxidation states has in

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aqueous solution.

(2)

Oxidation state of vanadium	Colour of the ions in aqueous solution
+2	mauve
+3	
+4	
+5	

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(ii) Two vanadium species exist in equilibrium in an aqueous solution.

$$VO_3^-(aq) + H_2O(l) \rightleftharpoons VO_2^+(aq) + 2OH^-(aq)$$

Deduce whether or not this is a redox reaction.
Justify your answer in terms of oxidation numbers.

(1)

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(iii) Predict the oxidising agent that could convert vanadium(II) ions to vanadium(III) ions and then vanadium(IV) ions, but **not** to vanadium(V) ions, under standard conditions.
Only use data from the tables shown.

Include equations and $E^{\ominus}_{cell}$ values for the two reactions that occur.

(5)

Vanadium systems

Electrode system	$E^{\ominus}$ / V
$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	−0.26
$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34
$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+1.00

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Oxidising agents

Electrode system	$E^{\ominus}$ / V
$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq)$	+0.15
$NO_3^-(aq) + 2H^+(aq) + e^- \rightleftharpoons NO_2(g) + H_2O(l)$	+0.80
$\frac{1}{2}Br_2(aq) + e^- \rightleftharpoons Br^-(aq)$	+1.09

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*(c) Transition metals, their compounds and their ions can act as heterogeneous and homogeneous catalysts.

Compare and contrast these two types of catalytic behaviour.
Include one specific example from transition metal chemistry of each of these types of catalyst and a reaction
in which it is used.

(6)

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(Total for question = 19 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Alanine is an amino acid.

(a) Which structure best represents alanine at **high** pH?

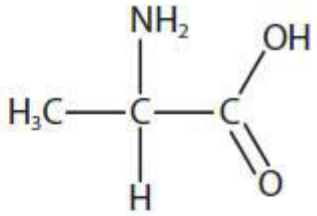
(1)

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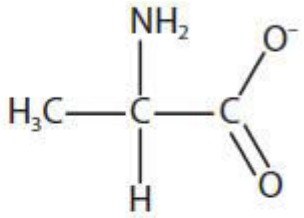
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☐ **A**

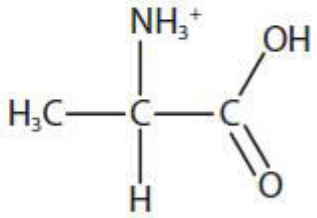


☐ **B**

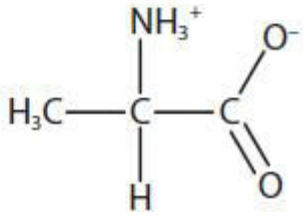


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☐ **C**



☐ **D**



(b) Alanine is a crystalline solid at room temperature.

What are the main forces broken when alanine melts?

(1)

☐ **A** London forces
☐ **B** hydrogen bonds
☐ **C** covalent bonds
☐ **D** ionic bonds

(Total for question = 2 marks)

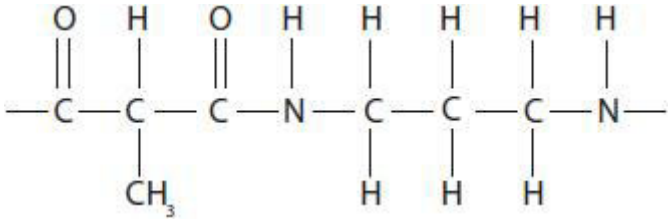
Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The repeat unit of a polymer is shown.

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This polymer could be

☒ **A** both a polypeptide and a polyamide

☒ **B** neither a polypeptide nor a polyamide

☒ **C** a polypeptide but not a polyamide

☒ **D** a polyamide but not a polypeptide

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Hydrogen bonds are formed when methylamine dissolves in water.

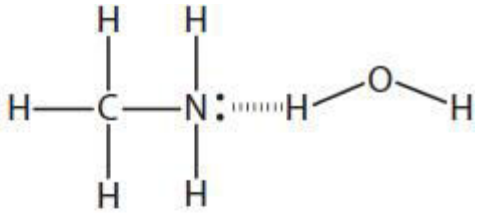
Which structure best represents a hydrogen bond between methylamine and water?

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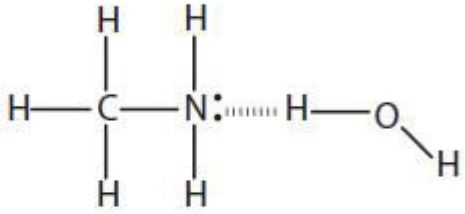
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☐ **A**

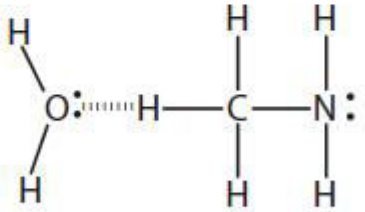


☐ **B**

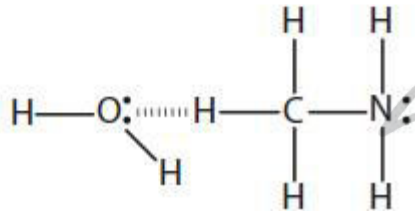


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☐ **C**



☐ **D**



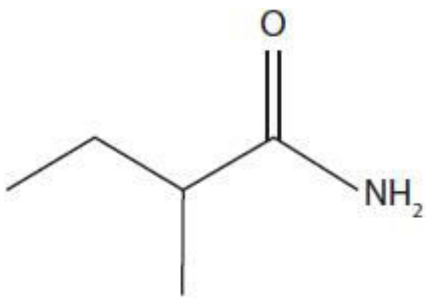
(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the name of the compound shown?

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☐ **A** 1-methylpropanamide

☐ **B** 3-methylpropanamide

☐ **C** 2-methylbutanamide

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☒ **D** 3-methylbutanamide

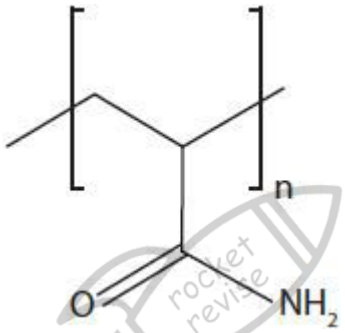
(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The repeat unit of a polymer is shown.

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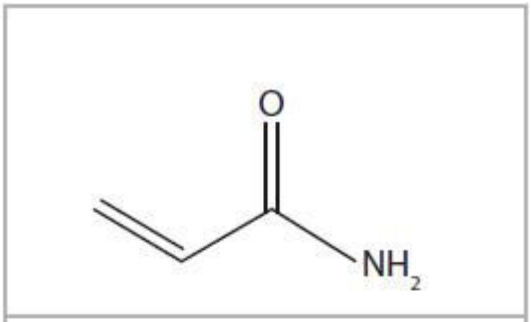
What is the structure of the monomer?

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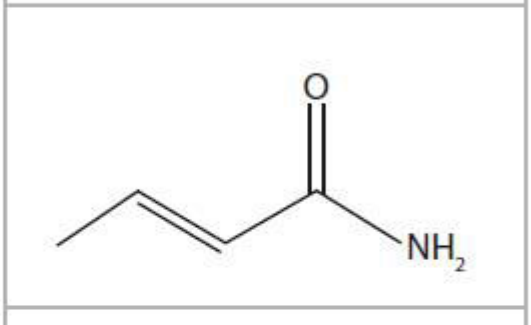
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☐ **A**

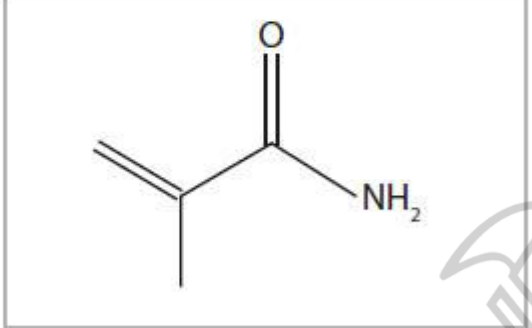


☐ **B**

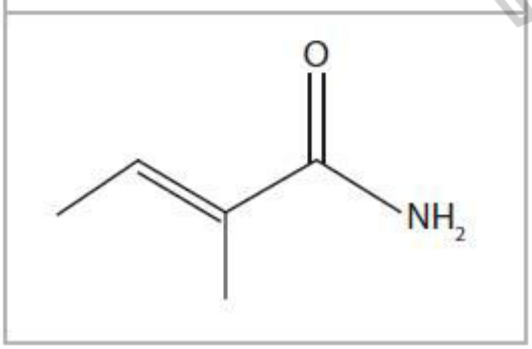


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☐ **C**



☐ **D**



(Total for question = 1 mark)

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Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Separate 0.1 mol dm^{-3} aqueous solutions of ammonia, butylamine and phenylamine were prepared.

Which of the following sequences shows the solutions in order of **increasing** pH?

☐ **A** butylamine, phenylamine, ammonia

☐ **B** ammonia, butylamine, phenylamine

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☒ **C** phenylamine, ammonia, butylamine

☒ **D** ammonia, phenylamine, butylamine

(Total for question = 1 mark)

Q12.

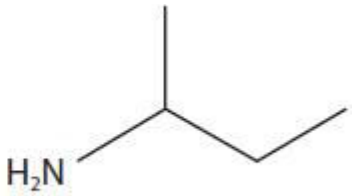
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which amine could **not** be prepared by the reduction of a nitrile?

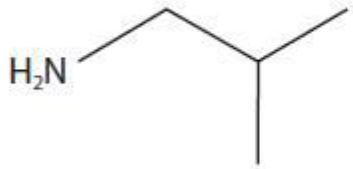
☒ **A**



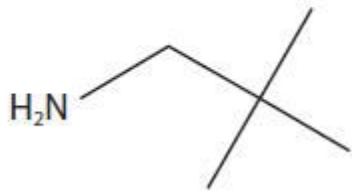
☒ **B**



☒ **C**



☒ **D**

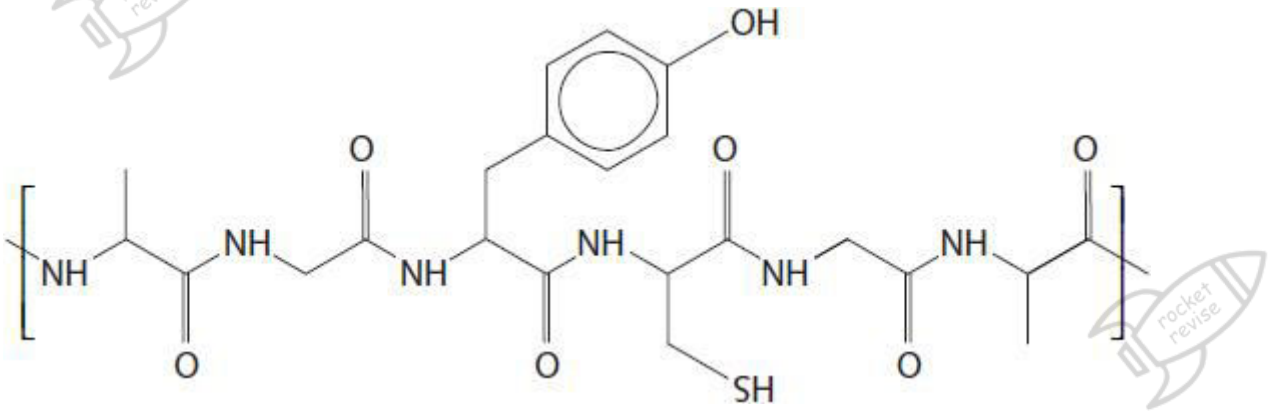


(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many **different** amino acids form the repeat unit of the polymer shown?



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- **A** 3
- **B** 4
- **C** 5
- **D** 6

(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which type of compound **cannot** be a monomer in the formation of polyamides?

- **A** amides
- **B** amino acids
- **C** diacyl chlorides
- **D** diamines

(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence shows these compounds in order of **decreasing** basicity?

- **A** $C_6H_5NH_2 > CH_3CH_2CH_2NH_2 > NH_3$
- **B** $NH_3 > CH_3CH_2CH_2NH_2 > C_6H_5NH_2$
- **C** $CH_3CH_2CH_2NH_2 > NH_3 > C_6H_5NH_2$
- **D** $C_6H_5NH_2 > NH_3 > CH_3CH_2CH_2NH_2$

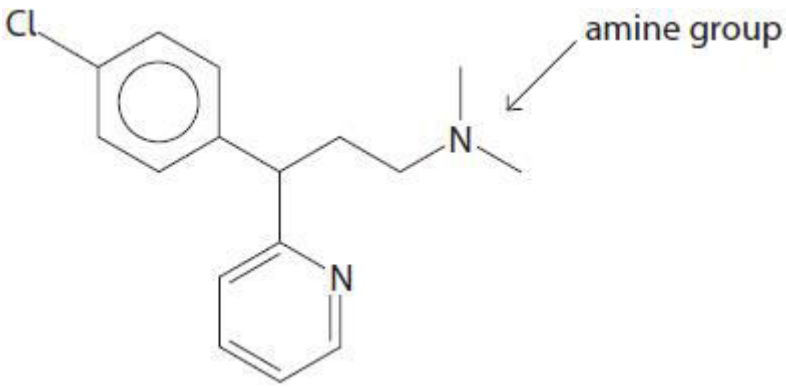
(Total for question = 1 mark)

Q16.

Chlorpheniramine is an amine used in the treatment of hayfever.

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(a) Deduce the classification of the labelled amine group.

(1)

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(b) The compound pyridine is used in the synthesis of chlorpheniramine.



pyridine

Like many amines, pyridine is miscible with water and the solution formed is alkaline.
Explain **each** of these properties of pyridine.

(4)

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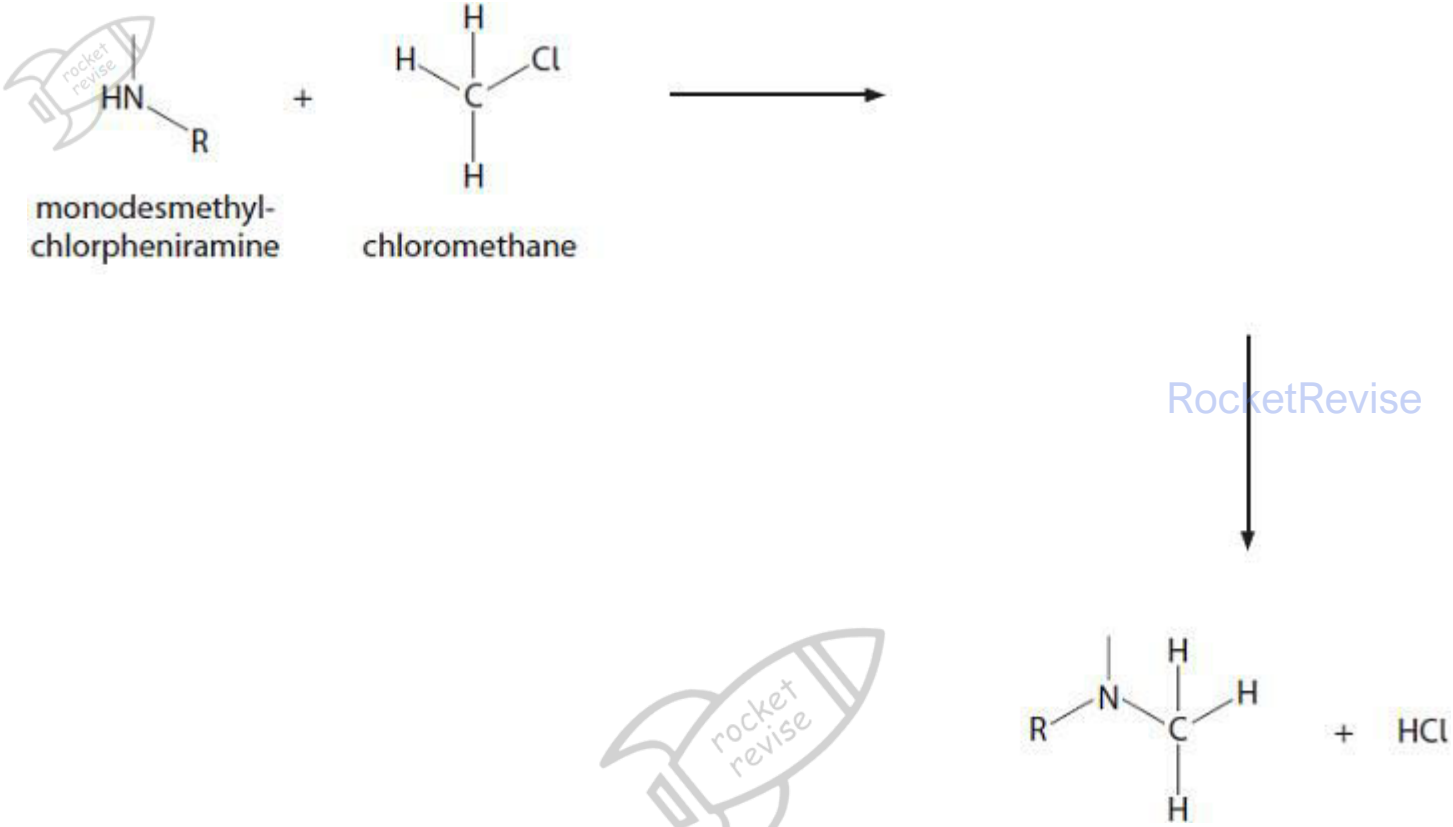
(c) A student suggested that the final step in the synthesis of chlorpheniramine starts with the reaction between monodesmethylchlorpheniramine and chloromethane.

Assuming the reaction is similar to that between ammonia and chloromethane, complete the mechanism for this proposed reaction.
Some of the organic structures shown have been simplified.
Include curly arrows, and any relevant dipoles and lone pairs of electrons.

(4)

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(Total for question = 9 marks)

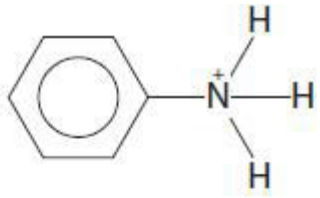
Q17.

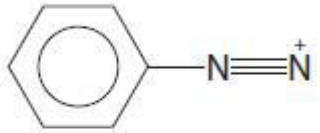
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which ion is formed when a mixture of sodium nitrite, $NaNO_2$, and dilute hydrochloric acid reacts with phenylamine at a temperature of 5°C?

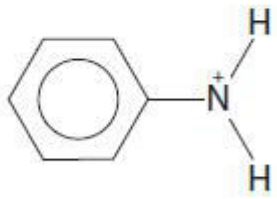
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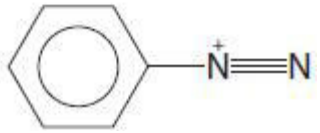
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☒ **A** 

☒ **B** 

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☒ **C** 

☒ **D** 

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(Total for question = 1 mark)

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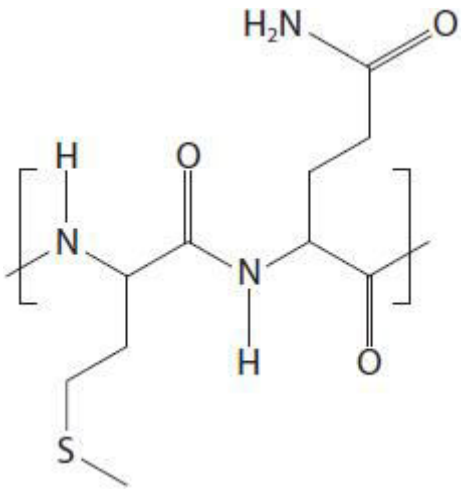
Q18.

This question is about polymers.

(a) The diagram shows part of the structure of a polymer formed by a **condensation** reaction between two amino acids.

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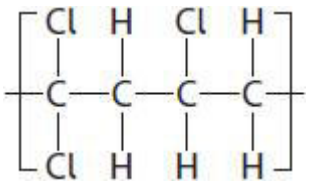


Predict the structures of the two monomers that produce this polymer.

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(2)

(b) The diagram shows a repeat unit of an addition polymer used in some food wraps. It is formed from two different monomers.



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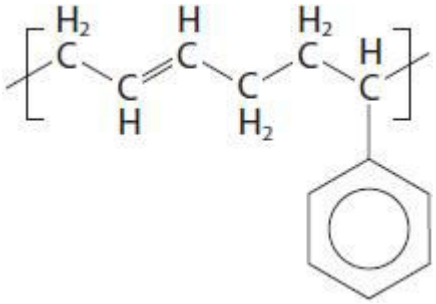
Deduce the structures of the two monomers that produce this polymer.

(2)

(c) A synthetic rubber polymer has the structure shown.

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The molar mass of the synthetic rubber is approximately 300 000 g mol^{-1}.
Calculate the approximate number of repeat units in the polymer.

(2)

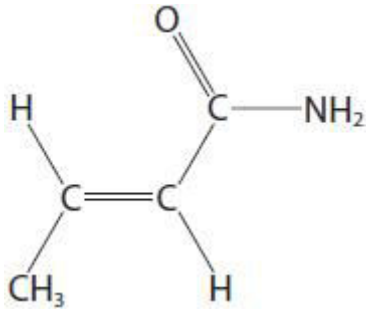
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(Total for question = 6 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of crotonamide is shown.



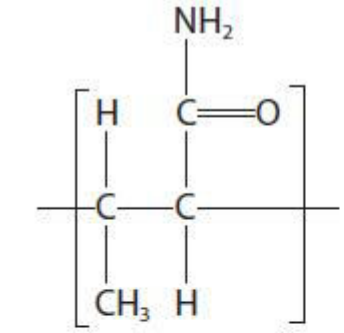
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What is the repeat unit of the polymer formed from crotonamide?

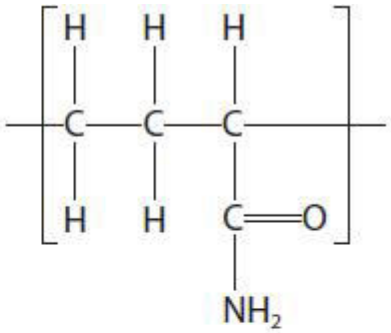
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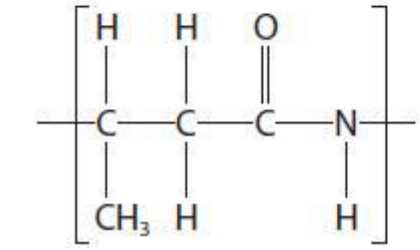
☐ **A**



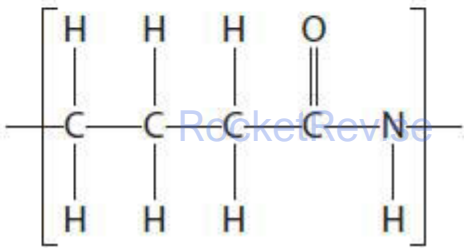
☐ **B**



☐ **C**



☐ **D**

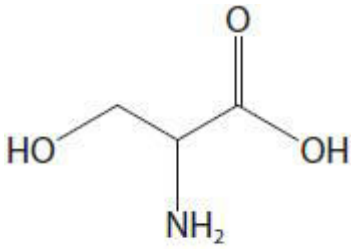


(Total for question = 1 mark)

Q20.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the amino acid serine.



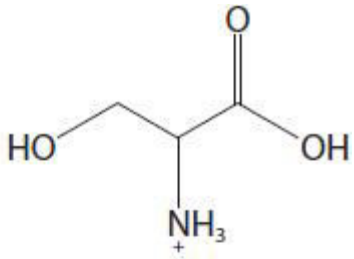
serine

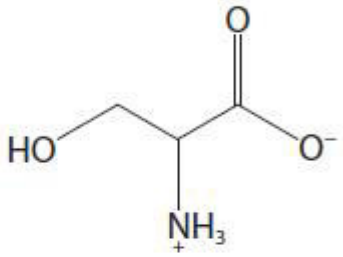
(a) Which ion is most likely to form if serine is dissolved in sodium carbonate solution?

(1)

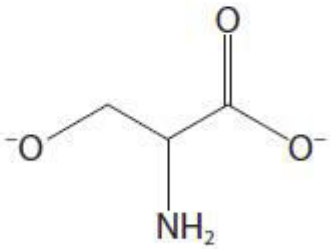
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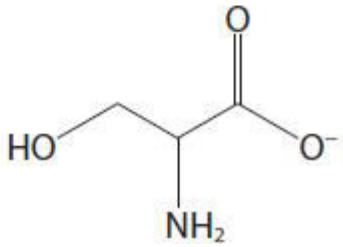
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☐ **A** 

☐ **B** 

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☐ **C** 

☐ **D** 

(b) Serine has a melting temperature in the region of 200–300 °C.

This high melting temperature is mainly due to the formation of

(1)

☐ **A** hydrogen bonds
☐ **B** ionic bonds
☐ **C** London forces
☐ **D** peptide bonds

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(Total for question = 2 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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When dilute hydrochloric acid is added to butylamine and the solution is allowed to evaporate to dryness, a white solid forms.

What is the formula of the white solid?

☒ **A** $CH_3CH_2CH_2CH_2Cl$
☒ **B** $CH_3CH_2CH_2CH_2NH_3Cl$
☒ **C** $CH_3CH_2CH_2CONH_2$
☒ **D** $CH_3CH_2CH_2COOH$

(Total for question = 1 mark)

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Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Amines may be prepared by the reduction of nitriles.

Identify the nitrile and the reducing agent used to prepare butylamine.

	Nitrile	Reducing agent
☐ **A**	propanenitrile	lithium tetrahydridoaluminate(III)
☐ **B**	propanenitrile	tin and concentrated hydrochloric acid
☐ **C**	butanenitrile	lithium tetrahydridoaluminate(III)
☐ **D**	butanenitrile	tin and concentrated hydrochloric acid

(Total for question = 1 mark)

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Q23.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The amino acid 2-aminopropanoic acid exists as a zwitterion at pH 6.0

What will be the structure of 2-aminopropanoic acid at pH 9.0?

- ☐ **A** $H_2NCH(CH_3)COOH$
- ☐ **B** $H_2NCH(CH_3)COO^-$
- ☐ **C** $H_3N^+CH(CH_3)COOH$
- ☐ **D** $H_3N^+CH(CH_3)COO^-$

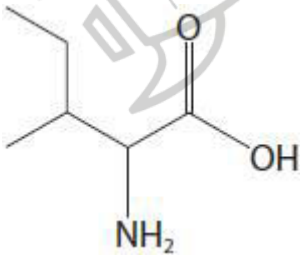
(Total for question = 1 mark)

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Q24.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of the amino acid isoleucine is shown.



(a) What is the systematic name of isoleucine?

(1)

- ☐ **A** 2-amino-3-ethylbutanoic acid
- ☐ **B** 2-amino-3-methylpentanoic acid
- ☐ **C** 3-amino-2-ethylbutanoic acid
- ☐ **D** 3-amino-2-methylpentanoic acid

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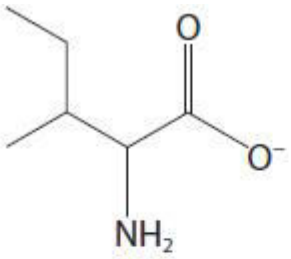
(b) What is the structure of isoleucine in a solution of pH = 2?

(1)

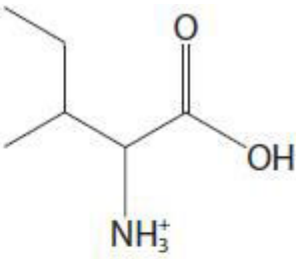
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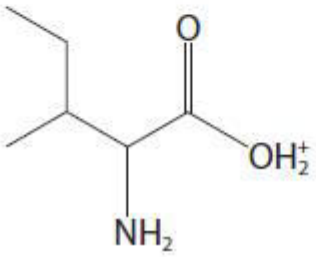
☐ **A**



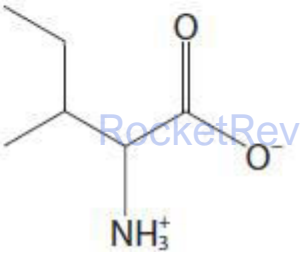
☐ **B**



☐ **C**



☐ **D**



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(Total for question = 2 marks)

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Q25.

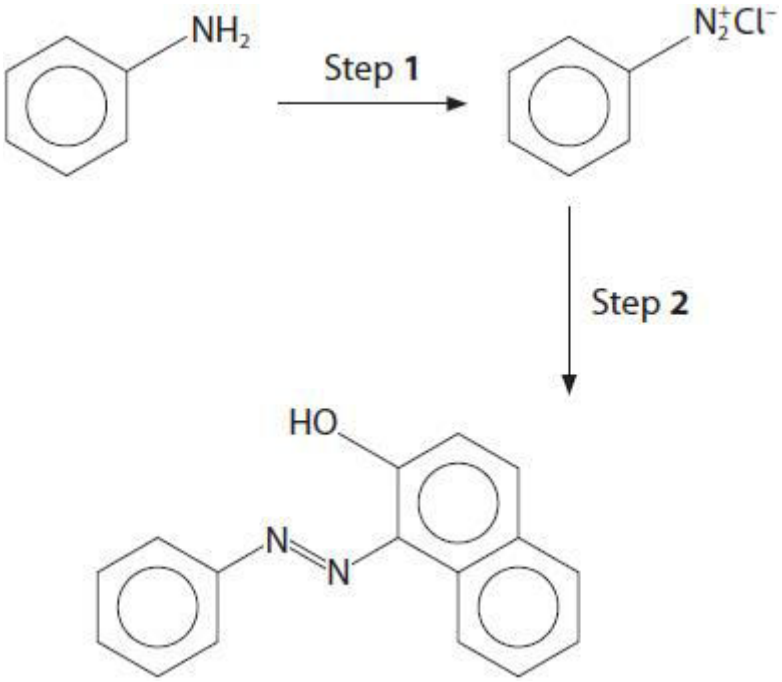
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A reaction scheme for the preparation of an azo dye from phenylamine is shown.

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(a) What are the reagents and condition for Step **1**?

(1)

- ☒ **A** $NaNO_2$ and HCl at 5 °C
- ☒ **B** $NaNO_3$ and HCl at 5 °C
- ☒ **C** $NaNO_2$ and HCl at 50 °C
- ☒ **D** $NaNO_3$ and HCl at 50 °C

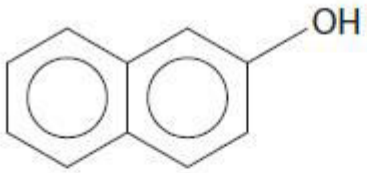
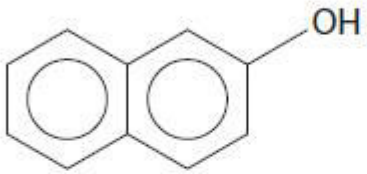
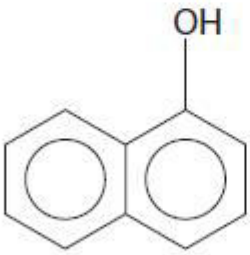
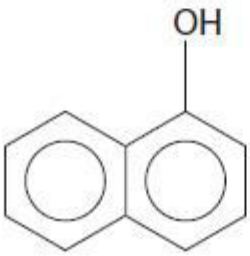
(b) Which reagent and condition are used for Step **2**?

(1)

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	Reagent	Condition
☐ A		acidic
☐ B		alkaline
☐ C		acidic
☐ D		alkaline

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(Total for question = 2 marks)

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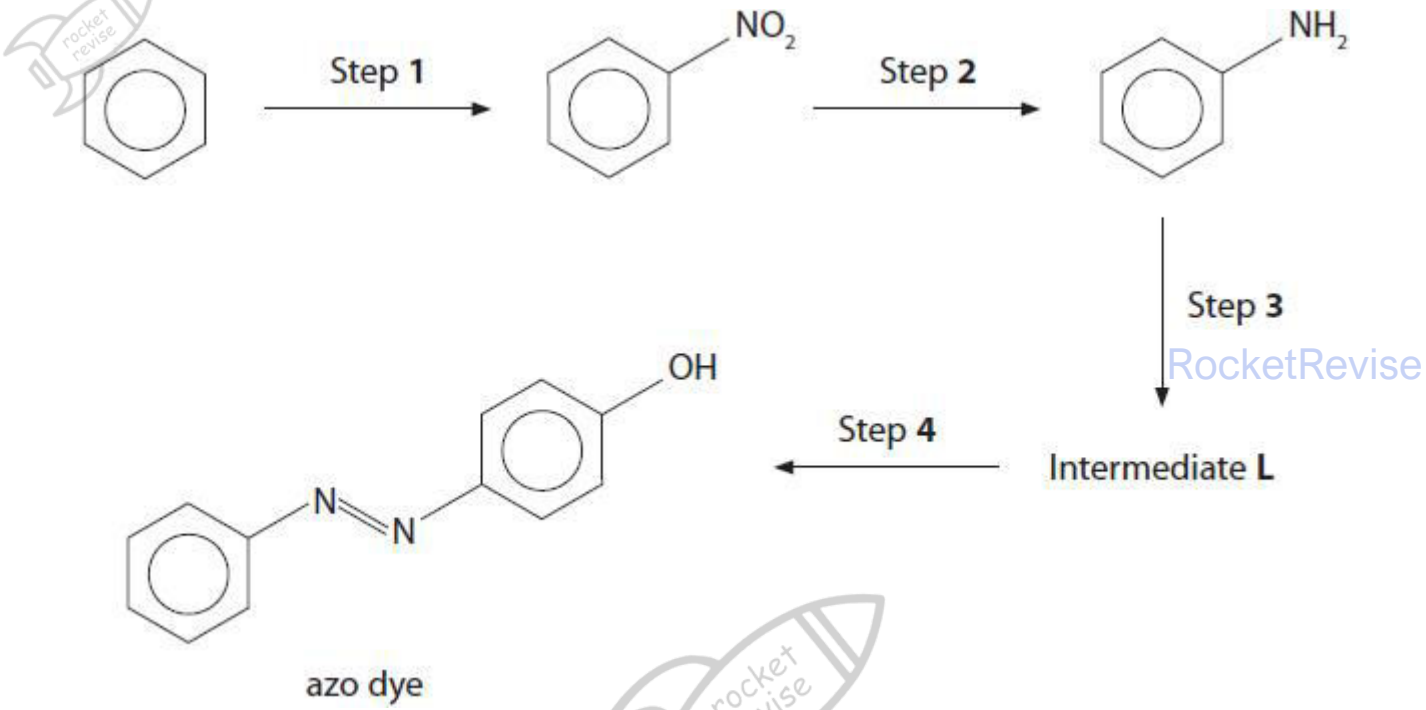
Q26.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A reaction scheme for the preparation of an azo dye is shown.



(a) Step **1** takes place using a mixture of concentrated nitric acid and concentrated sulfuric acid at 55°C.

Why is the reaction **not** carried out at 80°C?

(1)

- ☐ **A** further substitution by a nitro group occurs
- ☐ **B** nitrobenzene decomposes
- ☐ **C** substitution by SO_3H occurs
- ☐ **D** the nitric acid decomposes

(b) How is Step **2** carried out?

(1)

- ☐ **A** Sn, concentrated HCl(aq) and NaOH(aq) are added together
- ☐ **B** Sn, dilute HCl(aq) and NaOH(aq) are added together
- ☐ **C** Sn and concentrated HCl(aq) are added first, then NaOH(aq) is added at the end
- ☐ **D** Sn and dilute HCl(aq) are added first, then NaOH(aq) is added at the end

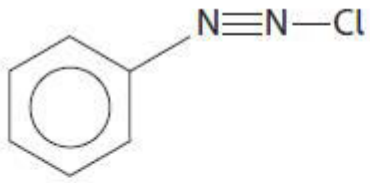
(c) Step **3** takes place using sodium nitrite and dilute hydrochloric acid at 5°C.

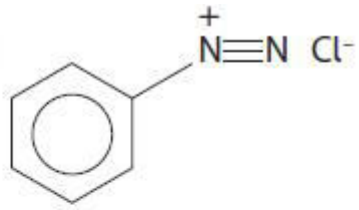
Which is the structure of Intermediate **L**?

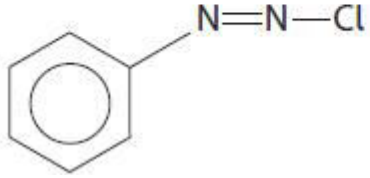
(1)

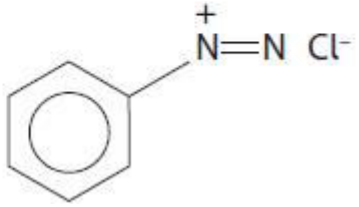
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☒ **A** 

☒ **B** 

☒ **C** 

☒ **D** 

(d) The reactant in Step **4** is phenol.

What condition is needed for this reaction?

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(1)

☒ **A** acidic
☒ **B** alkaline
☒ **C** ethanolic
☒ **D** neutral

(Total for question = 4 marks)

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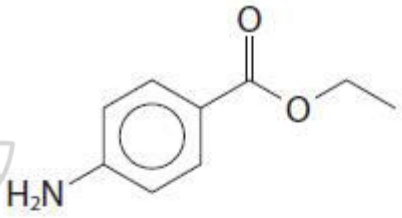
Q27.

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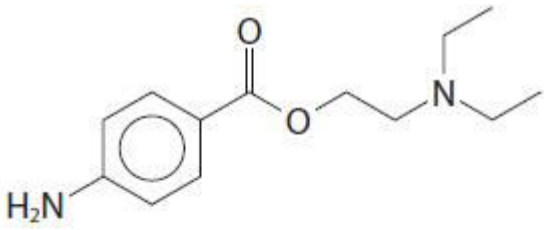
The synthetic drugs of the 'Caine' family are used as local anaesthetics. Caine drugs prevent nerve activity by binding to sodium channel receptors in the lipid cell membranes of neurons.

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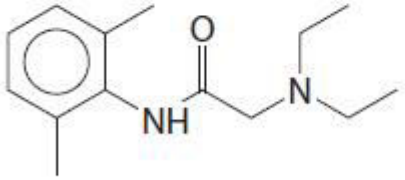
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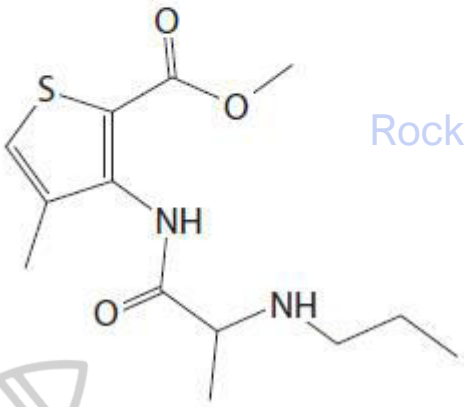
benzocaine



procaine



lidocaine



articaine

(a) Name the **three** functional groups present in benzocaine.

(2)

.....

.....

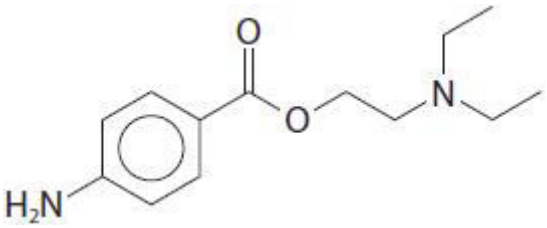
(b) Procaine can react with hydrochloric acid to form a monohydrochloride salt.

(i) Write an equation, using **molecular** formulae for this reaction. State symbols are not required.

(2)

(ii) Explain by considering **both** nitrogen atoms in procaine which nitrogen is more likely to be protonated in the reaction with hydrochloric acid.

(2)



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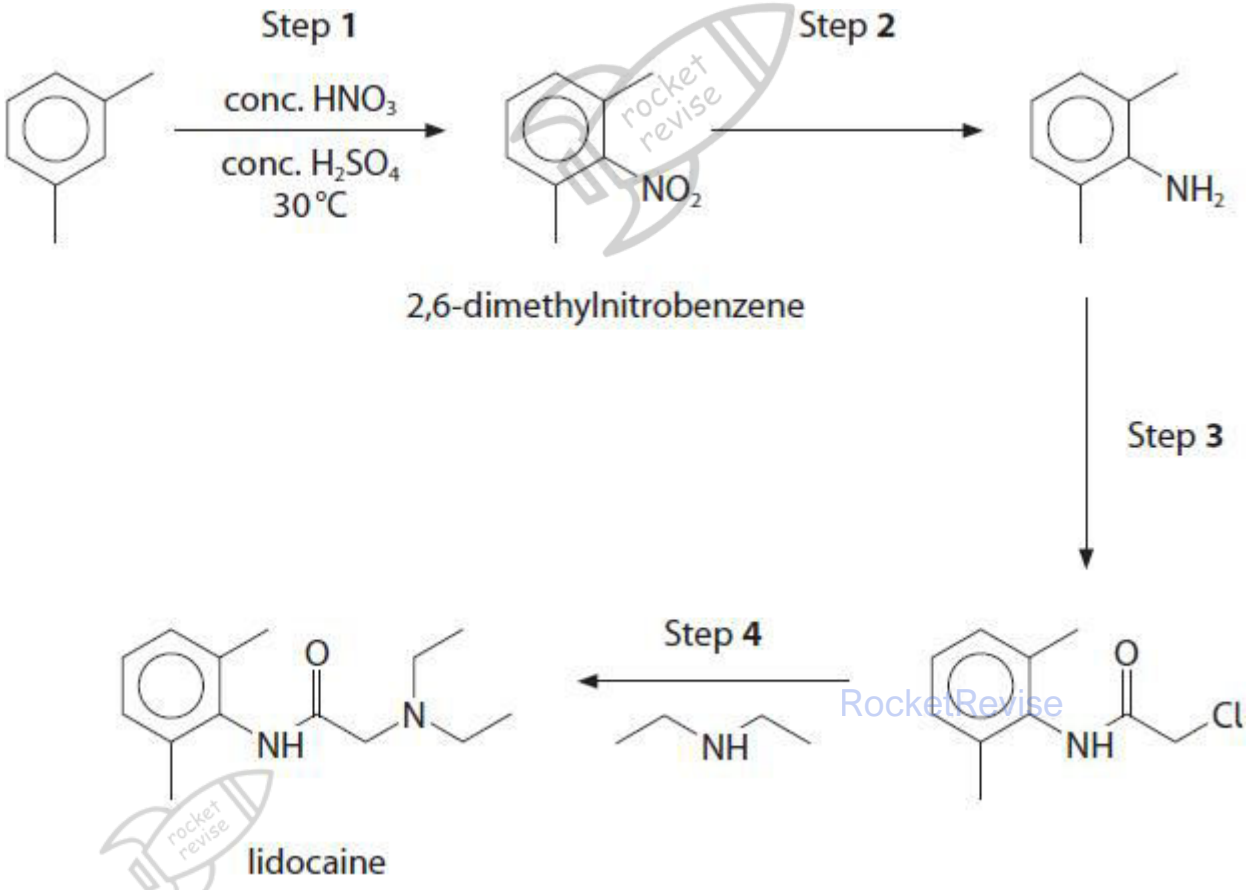
(c) When benzocaine and procaine are hydrolysed, one of the products formed is the same in both reactions.

Give the structure of this product.

(1)

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(d) Lidocaine can be prepared in the synthesis shown.



(i) Give the mechanism for Step **1**.
Include an equation for the formation of the electrophile.

(4)

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(ii) A low yield of 2,6-dimethylnitrobenzene is obtained in Step **1** due to the formation of additional organic products.

Give possible structures for **two** additional organic products.

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(2)

(iii) Give the reagents for Step **2**.

(1)

.....

.....

(iv) Deduce the **skeletal** formula of the reagent for Step **3**.

(1)

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(v) State the type and mechanism of reaction occurring in Step **4**.

(1)

.....

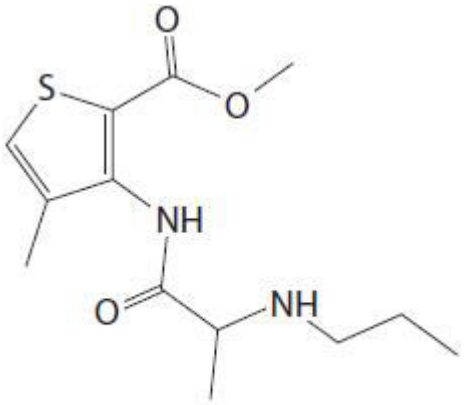
(e) Articaine is a chiral molecule.

Indicate any chiral centres on the structure of articaine.

(1)

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articaine

(f) Articaine is metabolised in the body with a half-life of 20 minutes.

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A patient is given a dose of 100 mg of articaine.
Calculate the mass of articaine **in micrograms**, μg, remaining in the body after **4 hours**.
[1 μg = 10^{-3} mg]

(3)

(Total for question = 20 marks)

Q28.

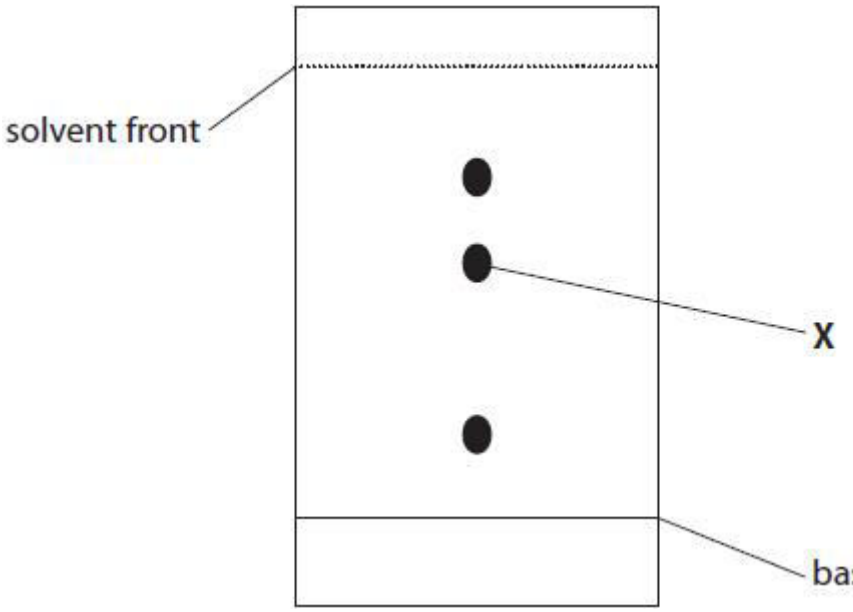
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student hydrolysed a section of a protein and then used thin-layer chromatography to analyse the amino acids produced.

The chromatogram is shown.

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What is the best estimate of the R_f value of the amino acid labelled **X**?

- ☒ **A** 0.2
- ☒ **B** 0.4
- ☒ **C** 0.6
- ☒ **D** 0.8

(Total for question = 1 mark)

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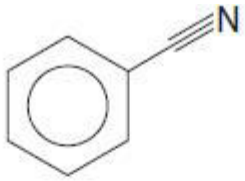
Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction can form a **secondary** amine?

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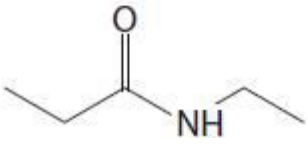
☐ **A** 

$\xrightarrow{LiAlH_4}$

☐ **B** $C_2H_5NH_2$ $\xrightarrow{C_2H_5Br}$

☐ **C** $(C_2H_5)_2NH$ $\xrightarrow{C_2H_5Br}$

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☐ **D** 

$\xrightarrow{NaOH}$

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

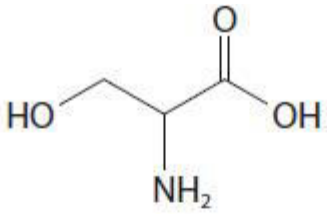
Four amino acids are dissolved in deionised water to form separate 0.05 mol dm^{-3} solutions.

Which amino acid will give the solution with the **lowest** pH?

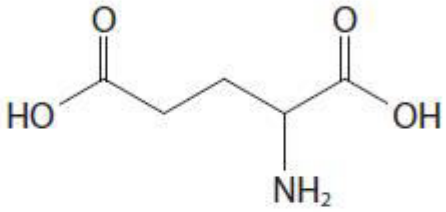
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☐ **A**

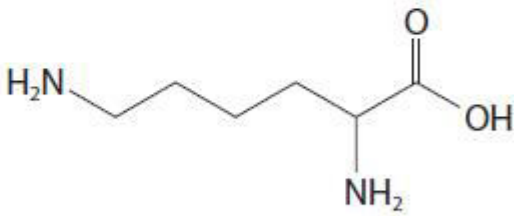


☐ **B**

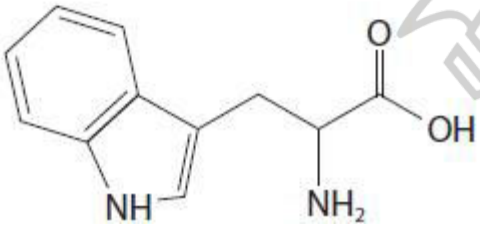


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☐ **C**



☐ **D**



(Total for question = 1 mark)

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Q31.

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Manganese and some of its compounds

Manganese is the twelfth most abundant element in the Earth's crust and it is an essential element for all living organisms.

Manganese is a transition metal and it can form compounds with oxidation states +2, +3, +4, +5, +6 and +7, although not all these are stable and some readily undergo disproportionation.

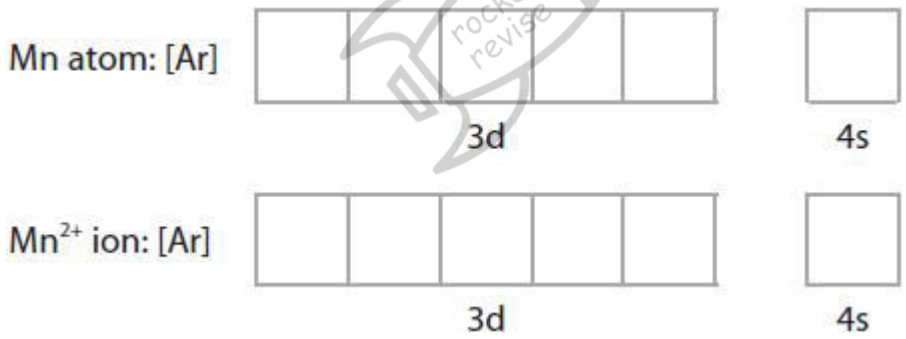
Some compounds and ions containing manganese are effective heterogeneous and homogeneous catalysts.

Alkaline batteries have a manganese(IV) oxide cathode and are used in digital cameras and toys. They have a similar voltage to traditional zinc-carbon batteries but have a higher energy density and last longer.

(a) Some of the properties of manganese and its compounds depend on electronic configurations.

(i) Complete the electronic configurations for a manganese atom and a manganese(II) ion, using the electron-in-boxes notation.

(1)



(ii) The standard electrode potentials, $E^{\ominus}$, for the half-cells involving the oxidation states +3 and +2 for manganese and iron are shown.

$$Mn^{3+}(aq) + e^- \rightleftharpoons Mn^{2+}(aq) \quad E^{\ominus} = +1.49V$$

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \quad E^{\ominus} = +0.77V$$

Explain, in terms of electronic configurations, why the $E^{\ominus}$ value for the Mn^{2+} / Mn^{3+} half-cell is significantly higher than that for the Fe^{2+} / Fe^{3+} half-cell.

(3)

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(b) Manganese(IV) oxide, MnO_2, is an oxidising agent and a catalyst.

(i) Chlorine can be prepared in a laboratory by reacting manganese(IV) oxide with concentrated hydrochloric acid.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$

Explain, in terms of oxidation numbers, why this is a redox reaction.

(2)

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(ii) Manganese(IV) oxide catalyses the decomposition of hydrogen peroxide.

$$2H_2O_2(aq) \rightarrow 2H_2O(aq) + O_2(g)$$

100 cm^3 of a solution of hydrogen peroxide decomposed to produce 86.0 cm^3 of oxygen gas, measured at room temperature and pressure (r.t.p.).
Calculate the concentration of the hydrogen peroxide solution in mol dm^{-3} .
[Molar volume of gas at r.t.p. = 24.0 dm^3 mol^{-1}]

(2)

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(c) Potassium manganate(VI), K_2MnO_4, is a green solid.
In acidic solution, the manganate(VI) ions disproportionate to form manganate(VII) ions and manganese(IV) oxide.

Write the **ionic** equation for this disproportionation.
State symbols are not required.

(1)

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(d) Manganate(VII) ions react with ethanedioate ions in acidic solution.

$$2MnO_4^-(aq) + 16H^+(aq) + 5C_2O_4^{2-}(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10CO_2(g)$$

Two experiments were carried out:

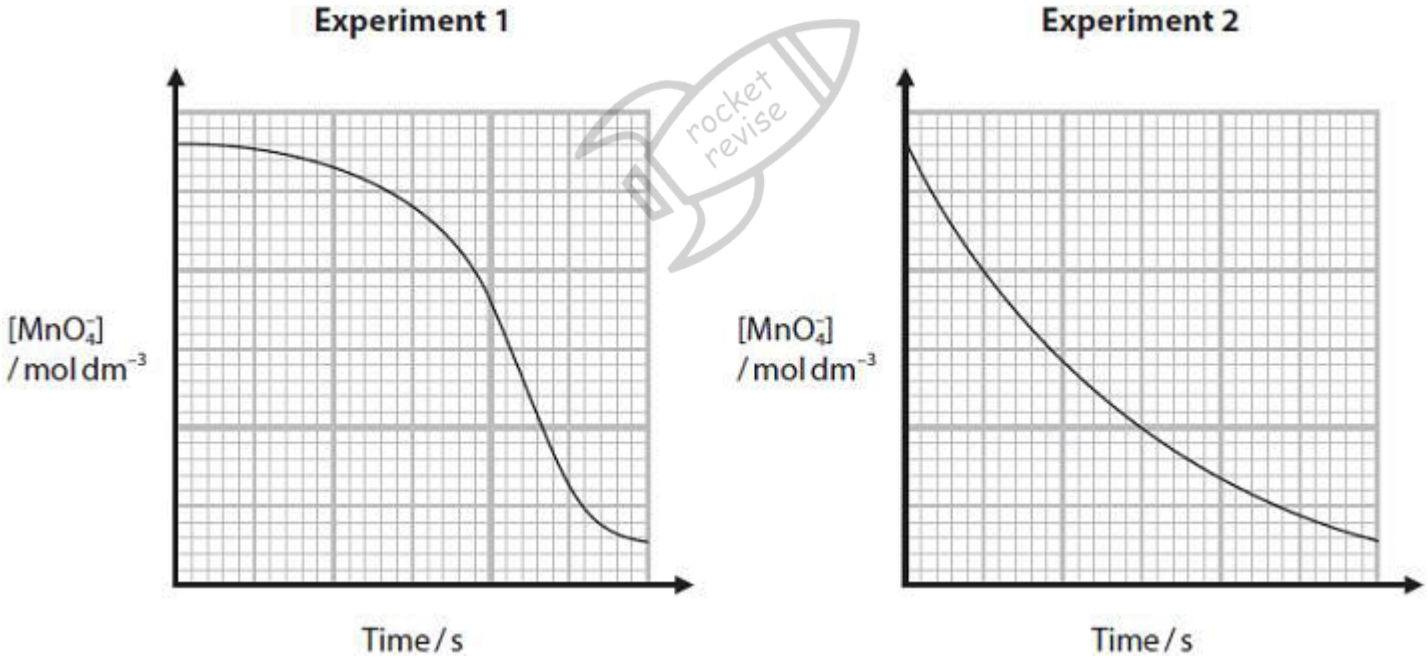
Experiment 1

Aqueous potassium manganate(VII) was added to a mixture of excess ethanedioic acid and sulfuric acid. The concentration of the manganate(VII) ions was determined as the reaction progressed.

Experiment 2

The same volumes and concentrations of solutions as in **Experiment 1** were used but some solid manganese(II) sulfate was dissolved in the ethanedioic acid and sulfuric acid mixture before adding the potassium manganate(VII) solution.

Graphs of the concentration of manganate(VII) ions against time are shown.



Explain, in terms of catalysis, why the graphs have different shapes.

(3)

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(e) Methylbenzene is oxidised to benzoic acid by heating under reflux with an alkaline solution of potassium manganate(VII).

In an experiment, 1.73 g of methylbenzene was mixed with 7.00 g of potassium manganate(VII) and excess potassium hydroxide solution.
The overall equation is

$$2MnO_4^- + C_6H_5CH_3 \rightarrow 2MnO_2 + C_6H_5COO^- + OH^- + H_2O$$

(i) Show that the potassium manganate(VII) is in excess.
[Molar masses / g mol^{-1}: $C_6H_5CH_3$ = 92 $KMnO_4$ = 158]

(3)

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(ii) State how the benzoate ions, $C_6H_5COO^-$, can be converted into benzoic acid.

(1)

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(f) Alkaline batteries consist of an anode made of a zinc alloy, a cathode made of manganese(IV) oxide and an electrolyte of aqueous potassium hydroxide.

The overall equation for the reaction in the cell is

$$2MnO_2(s) + Zn(s) \rightleftharpoons Mn_2O_3(s) + ZnO(s) \qquad E^{\ominus}_{cell} = +1.43\,V$$

(i) Write the half-equations for the reactions at the anode and the cathode.

(2)

anode half-equation:

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cathode half-equation:

(ii) The standard electrode potential for the Zn / ZnO half-cell is –1.28 V.
Calculate the standard electrode potential for the MnO_2 / Mn_2O_3 half-cell.

(1)

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(Total for question = 19 marks)

Q32.

This question is about some organic nitrogen compounds.

(a) Phenol reacts with dilute nitric acid at room temperature to form a mixture of 2-nitrophenol and 4-nitrophenol.

The reaction is an electrophilic substitution.

(i) Draw a mechanism for the reaction to form 2-nitrophenol.

Assume that NO_2^+ is the electrophile.

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(3)

(ii) Benzene requires a mixture of concentrated nitric acid and concentrated sulfuric acid at 60 °C to form nitrobenzene.

Explain why phenol requires milder conditions for nitration than benzene.

(2)

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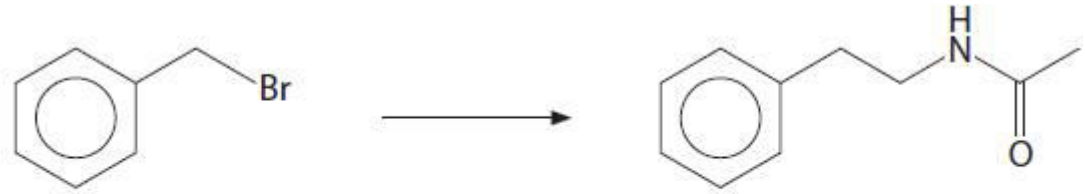
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(b) Devise a synthesis of N-phenylethylethanamide from (bromomethyl)benzene **without** the use of a Grignard reagent.

Include reagents, conditions and the structures of the intermediate compounds.

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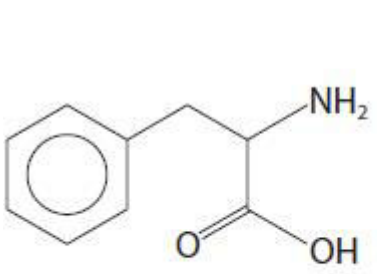
(bromomethyl)benzene

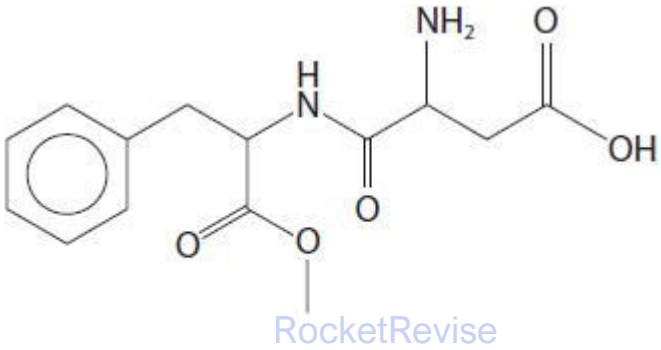
N-phenylethylethanamide

(5)

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(c) Phenylalanine is a naturally-occurring amino acid found in meat, milk and eggs. It is used to form the artificial sweetener known as aspartame.





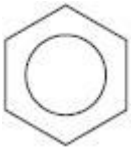
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phenylalanine

aspartame

(i) Complete the structure of the zwitterion of phenylalanine.

(1)



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(ii) Draw the structure of the dipeptide formed from phenylalanine.

(1)

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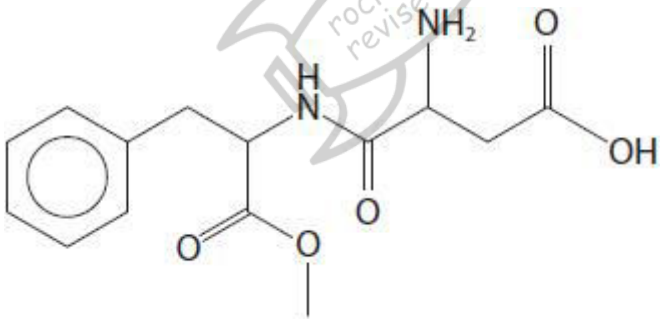
(iii) Give the molecular formula of aspartame.

(1)

.....

(iv) Circle the **two** chiral carbon atoms in aspartame on the structure shown.

(1)



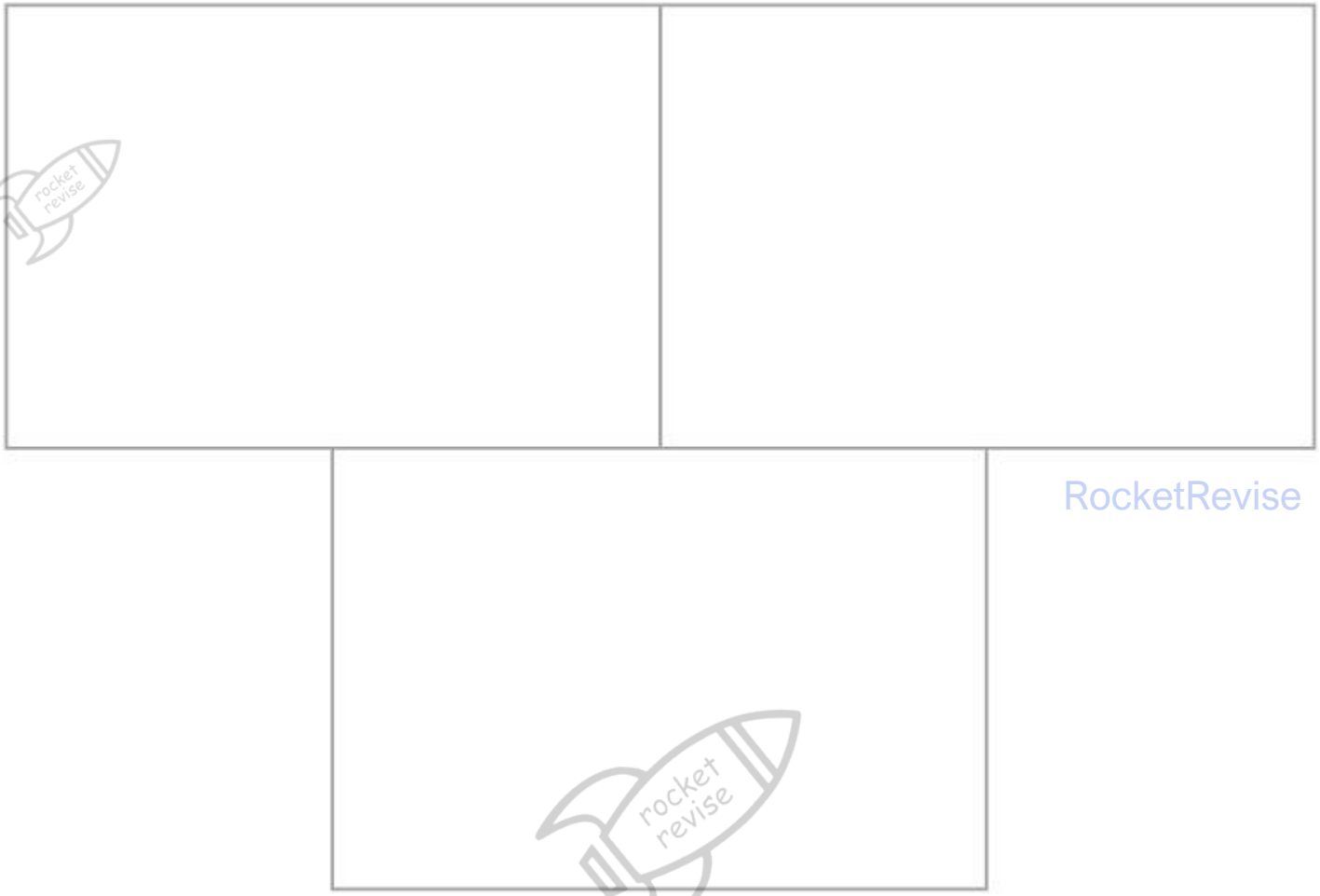
(v) Aspartame is hydrolysed by hydrochloric acid in the stomach.
Draw the structures of the three organic products formed from the complete acid hydrolysis of aspartame.

(3)

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(vi) A can of a fizzy drink contains 53 mg of aspartame per 100 cm^3 of drink.
Calculate the number of molecules of aspartame in a 330 cm^3 can of the drink.
Give your answer to an appropriate number of significant figures.
[Molar mass aspartame = 294 g mol^{-1}
Avogadro constant = 6.02×10^{23} mol^{-1}]

(3)

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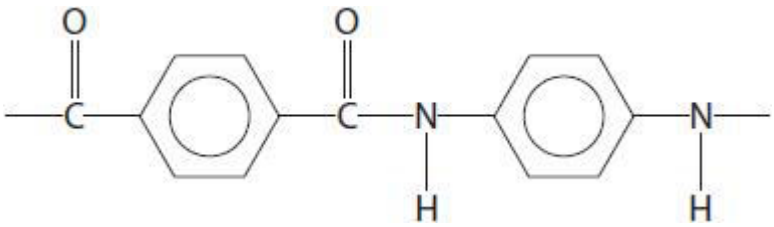
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(Total for question = 20 marks)

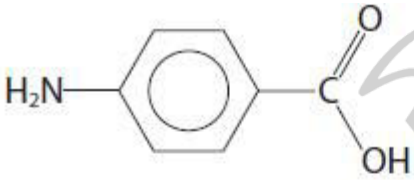
Q33.

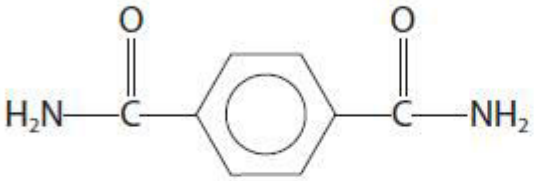
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒

A polyamide has the repeat unit



Which of these monomers can form this polyamide?

☐ **A**  only

☐ **B**  only

☐ **C**  and 

☐ **D**  and 

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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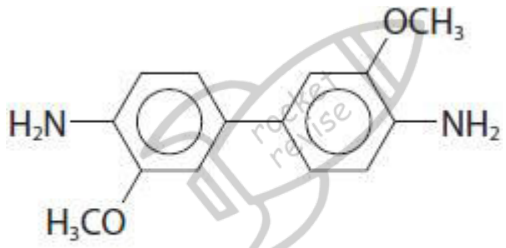
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Q34.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Compound **X** reacts with **excess** nitrous acid, followed by coupling with **excess** phenol to form an azo dye.



compound **X**

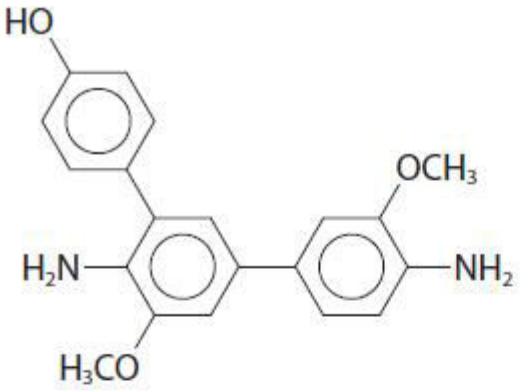
What is the structure of the azo dye?

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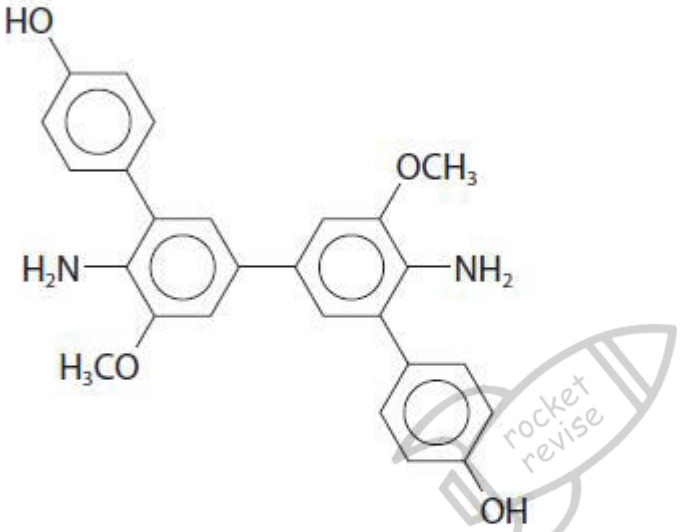
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☐ **A**

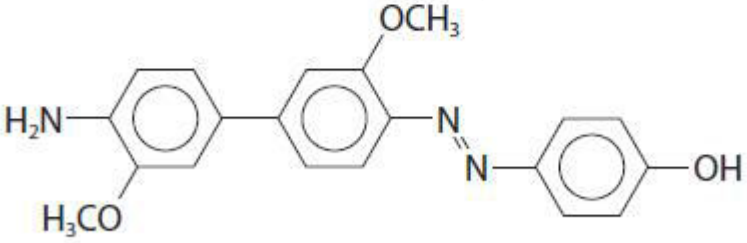


☐ **B**

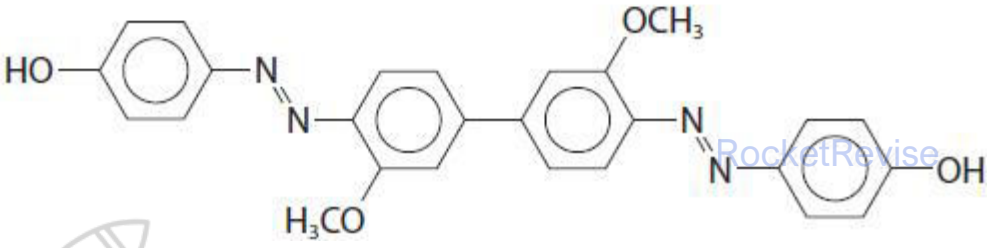


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☐ **C**



☐ **D**



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(Total for question = 1 mark)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which of these is the **weakest** base?

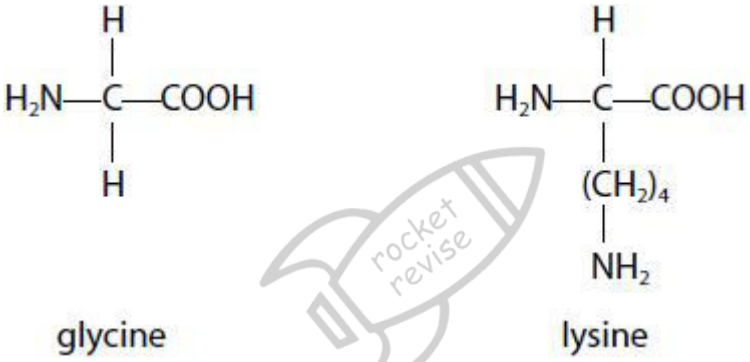
- **A** ammonia
- **B** butylamine
- **C** ethylamine
- **D** phenylamine

(Total for question = 1 mark)

Q36.

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Glycine and lysine are two naturally-occurring amino acids.



(a) Write the equation for the reaction of glycine with sodium hydroxide.

State symbols are not required.

(1)

(b) Calculate the volume, in cm^3, of 0.100 mol dm^{-3} hydrochloric acid required to completely react with 1.825 g of lysine.

[M_r of lysine = 146]

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(2)

(c) Lysine exists as optically active enantiomers but glycine does not.

(i) Draw three-dimensional diagrams of the two optically active lysine enantiomers.

(2)

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(ii) Describe how these optically active enantiomers could be distinguished.
Practical details are not required.

(2)

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(iii) State why glycine does not exist as enantiomers.

(1)

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(d) Chromatography can be used to separate a mixture of glycine and lysine.

Draw spots to show the location of glycine and lysine on the chromatogram, given that their R_f values are 0.26 and 0.14 respectively.

(1)

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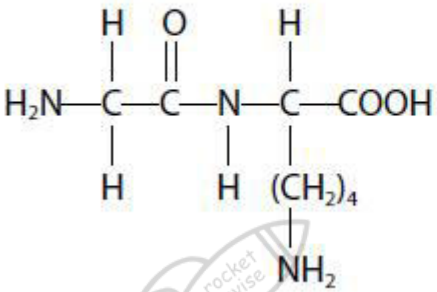


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(e) Naturally-occurring glycine and lysine can join together to form different dipeptides.

Draw a different dipeptide of glycine and lysine.

(1)

Dipeptide 1	Dipeptide 2
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(Total for question = 10 marks)

Q37.

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Identify the correct trend of **increasing** strength as a base.

(1)

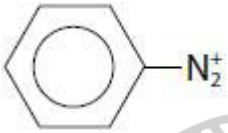
- ☒ **A** $C_6H_5–NH_2 < H–NH_2 < CH_3–NH_2$
- ☒ **B** $C_6H_5–NH_2 < CH_3–NH_2 < H–NH_2$
- ☒ **C** $H–NH_2 < CH_3–NH_2 < C_6H_5–NH_2$
- ☒ **D** $H–NH_2 < C_6H_5–NH_2 < CH_3–NH_2$

(Total for question = 1 mark)

Q38.

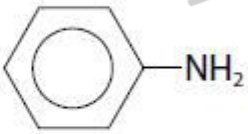
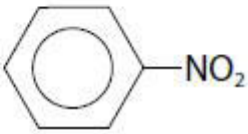
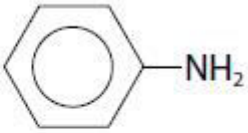
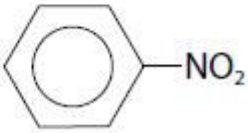
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Azo dyes are made from the benzenediazonium ion.



(a) Benzenediazonium ions can be made from:

(1)

	Reagent 1	Reagent 2
☐ **A**	HNO_2	
☐ **B**	HNO_2	
☐ **C**	HNO_3	
☐ **D**	HNO_3	

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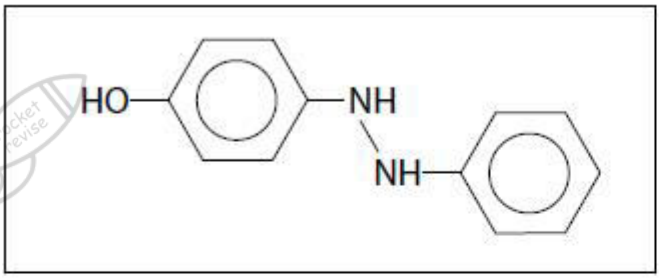
(b) The structure of the azo dye formed when benzenediazonium ions react with phenol is

(1)

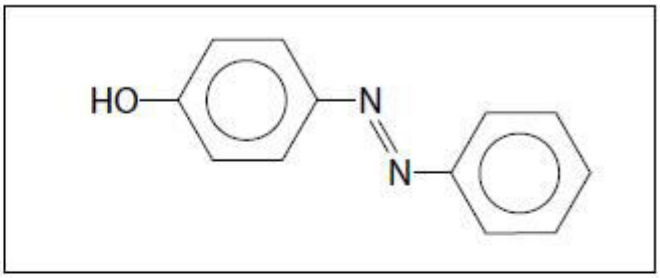
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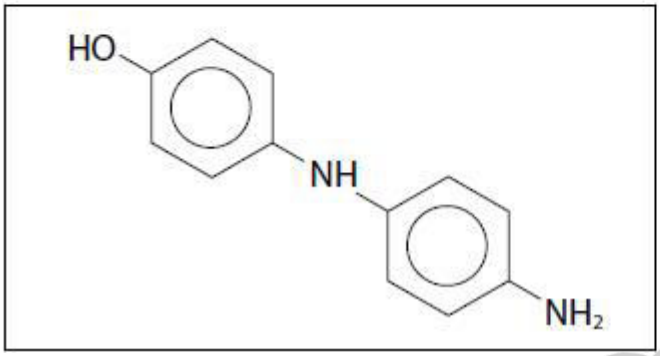
☐ **A**



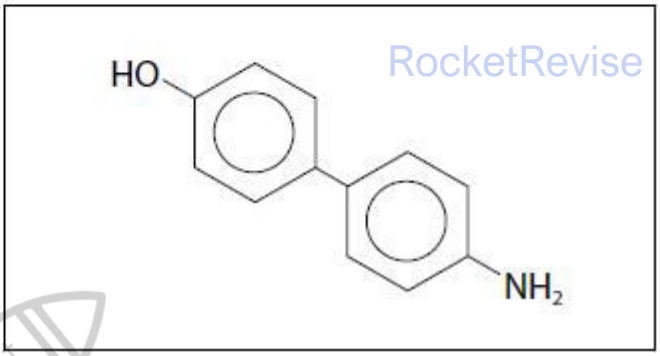
☐ **B**



☐ **C**



☐ **D**



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(Total for question = 2 marks)

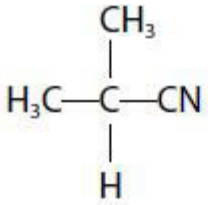
Q39.

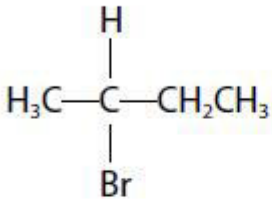
Which combination of reactants will produce a primary aliphatic amine as the product?

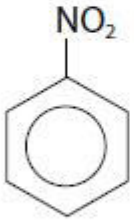
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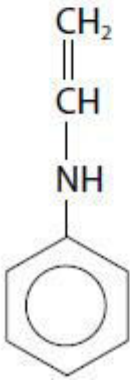
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☐ **A**  + Ni/H_2

☐ **B**  + CH_3NH_2

☐ **C**  + Sn + conc HCl

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☐ **D**  + Ni/H_2

(Total for question = 1 mark)

Q40.

Which equation shows the two compounds that react to produce ethanamide, CH_3CONH_2, in a single step?

(1)

☐ **A** $CH_4 + HCONH_2 \rightarrow CH_3CONH_2 + H_2$

☐ **B** $CH_3COOH + NH_3 \rightarrow CH_3CONH_2 + H_2O$

☐ **C** $CH_3COCl + NH_3 \rightarrow CH_3CONH_2 + HCl$

☐ **D** $CH_3CHO + NH_3 \rightarrow CH_3CONH_2 + H_2$

(Total for question = 1 mark)

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Q41.

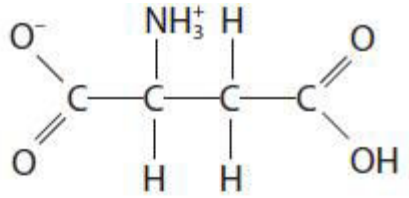
Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Aspartic acid is a naturally-occurring amino acid.

(a) Which form of aspartic acid is formed at pH 12?

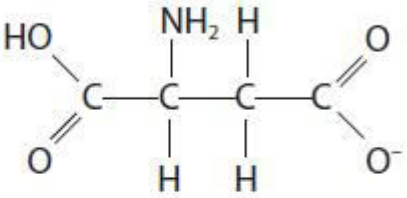
(1)

☐ **A**

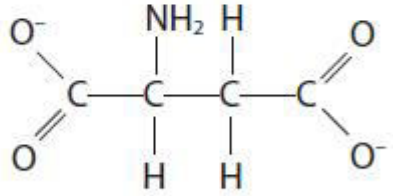


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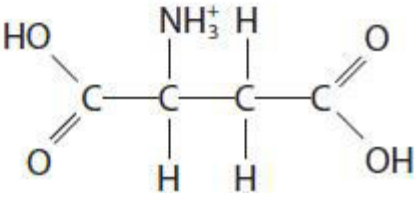
☐ **B**



☐ **C**

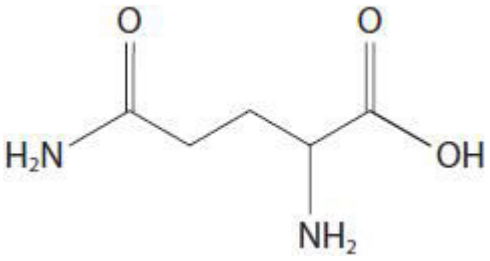


☐ **D**



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(b) Glutamine is shown.



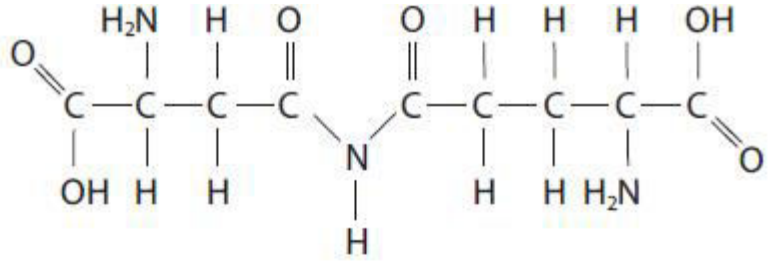
(i) Which is a correct structure when aspartic acid forms a dipeptide with glutamine?

(1)

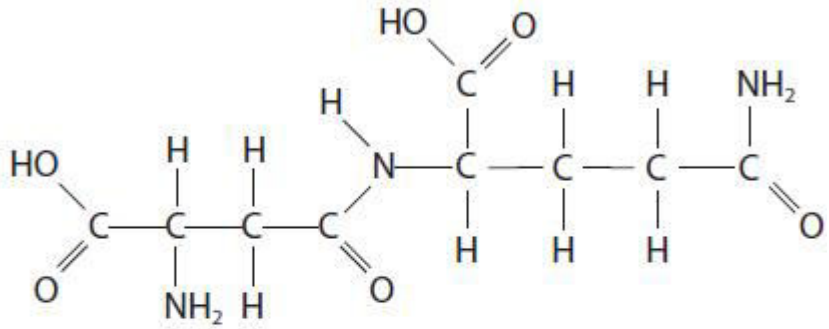
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☐ **A**

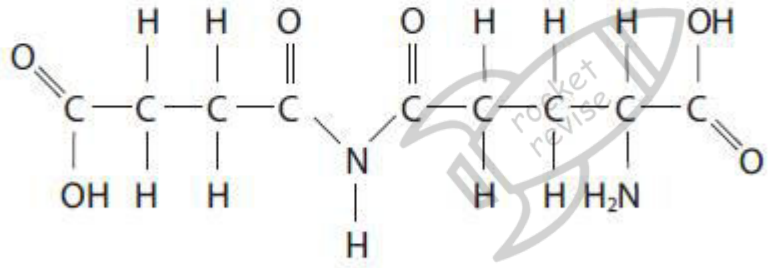


☐ **B**

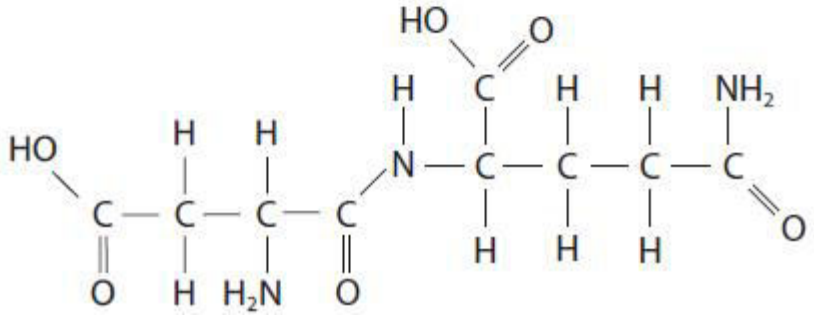


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☐ **C**



☐ **D**



(ii) What is the maximum number of chiral centres in the dipeptides in (b)(i)?

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(1)

☐ **A** 0
☐ **B** 1
☐ **C** 2
☐ **D** 3

(Total for question = 3 marks)

Q42.

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This question is about butylamine, $C_4H_9NH_2$.

The boiling temperature of butylamine is 78°C, which is much higher than alkanes of similar molar mass.

(a) Butylamine is miscible with water. Draw a labelled diagram to show one of the intermolecular interactions between a butylamine molecule and a molecule of water.

Include relevant lone pairs and dipoles.

(2)

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(b) Butylamine has a fishy, ammonia-like smell. The exposure limit or safety level in the atmosphere for butylamine is 15 mg m^{-3}.

Calculate the maximum number of butylamine molecules that are allowed per cubic metre.
[Avogadro constant, $L = 6.02 \times 10^{23}$ mol^{-1}]

(2)

(c) Butylamine reacts with ethanoyl chloride to form *N*-butylethanamide.

Write an equation for this reaction using **displayed** formulae.

(2)

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(d) Explain why butylamine is a stronger base than ammonia.

(3)

.....

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(Total for question = 9 marks)

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Q43.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A dipeptide has the molecular formula $C_7H_{12}N_2O_3$.

The dipeptide is hydrolysed to form two amino acids. One of the amino acids produced does not have a chiral centre.

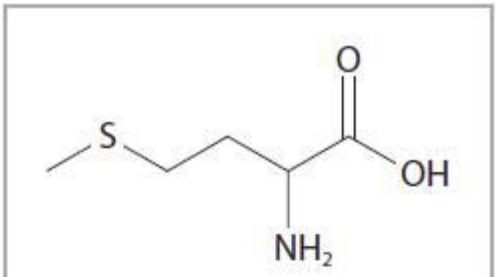
What is the structure of the other amino acid which does have a chiral centre?

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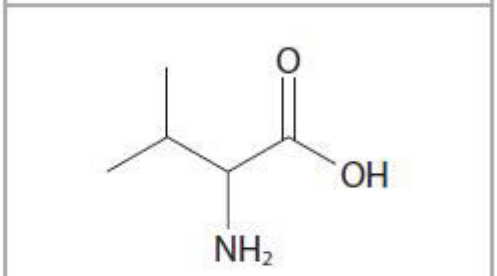
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☐ **A**

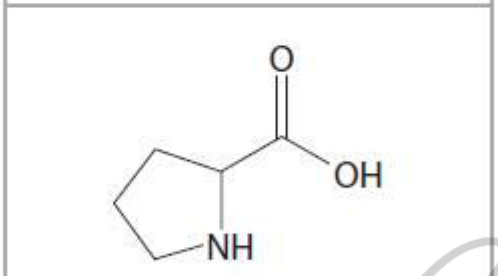


☐ **B**

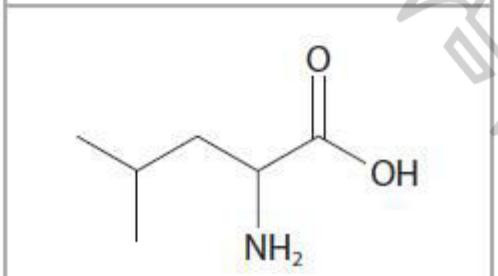


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☐ **C**



☐ **D**



(Total for question = 1 mark)

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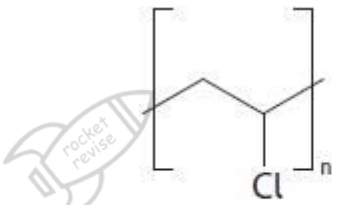
Q44.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

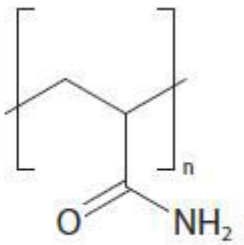
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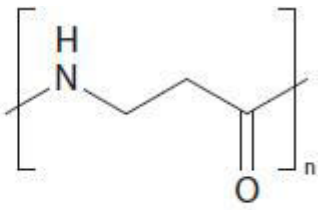
The formulae of three synthetic polymers, **X**, **Y** and **Z**, are shown.



X



Y



Z

Which are made by addition polymerisation reactions?

- ☒ **A** **X**, **Y** and **Z**
- ☒ **B** **X** and **Y**
- ☒ **C** **X** and **Z**
- ☒ **D** **Y** and **Z**

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(Total for question = 1 mark)

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Q45.

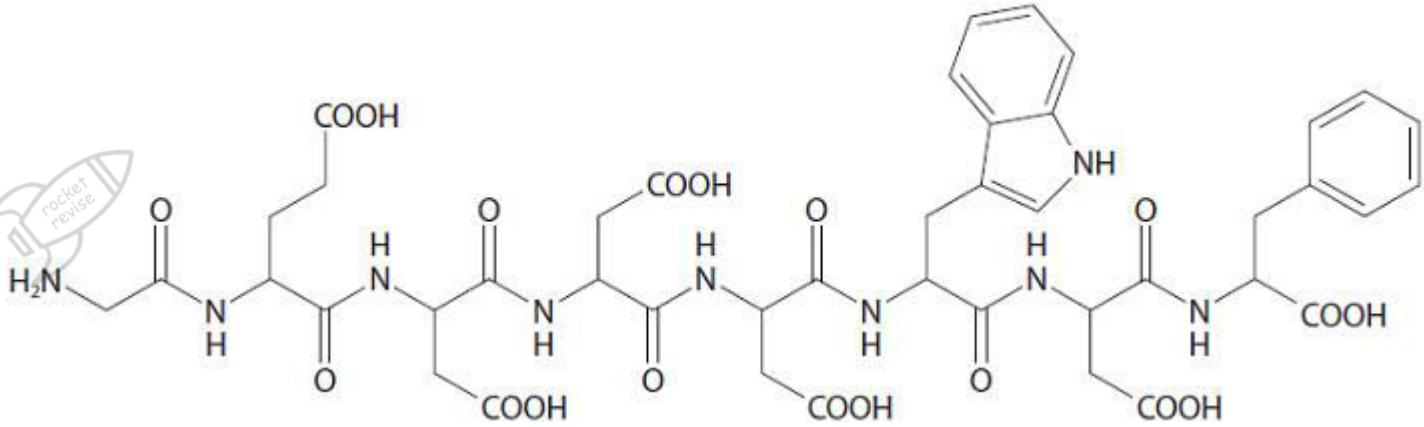
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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Peptides are short chains of amino acids linked by peptide bonds.

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How many **different** types of amino acid have joined to form the octapeptide?

- ☒ **A** 4
- ☒ **B** 5
- ☒ **C** 6
- ☒ **D** 8

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(Total for question = 1 mark)

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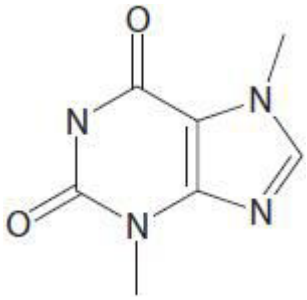
Q46.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the skeletal structure of theobromine, which has a bitter taste and is found in chocolate and tea leaves.

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Which of the functional groups listed is **not** present in the structure?

- ☒ **A** alkyl
- ☒ **B** amide
- ☒ **C** amine
- ☒ **D** ketone

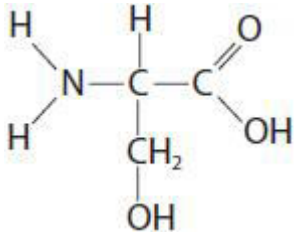
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(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The amino acid serine has the structure shown.



(a) The IUPAC name for serine is

(1)

- ☒ **A** 2-amino-2-carboxyethanol
- ☒ **B** 2-amino-1-hydroxyethanoic acid
- ☒ **C** 2-amino-1-hydroxypropanoic acid
- ☒ **D** 2-amino-3-hydroxypropanoic acid

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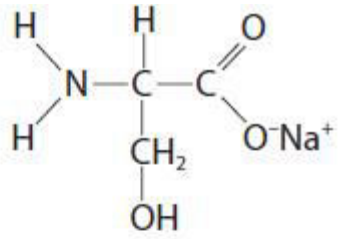
(b) Which is the organic product of the reaction between serine and sodium hydroxide solution?

(1)

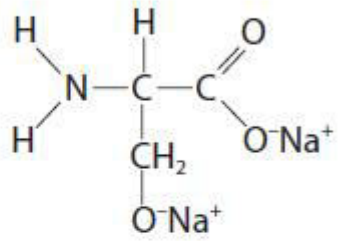
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☐ **A**

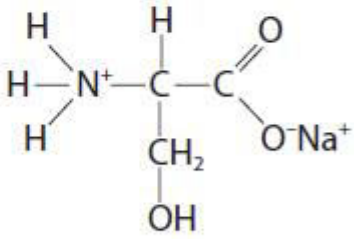


☐ **B**

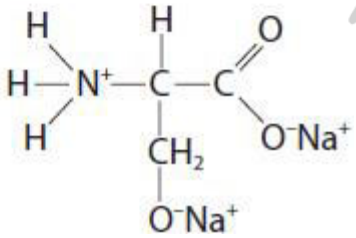


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☐ **C**



☐ **D**



(Total for question = 2 marks)

Q48.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

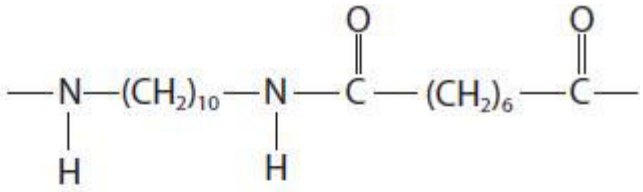
The polymer, nylon 6-10, is made from a six-carbon diamine and a ten-carbon dicarboxylic acid.

Which is the repeat unit of this nylon?

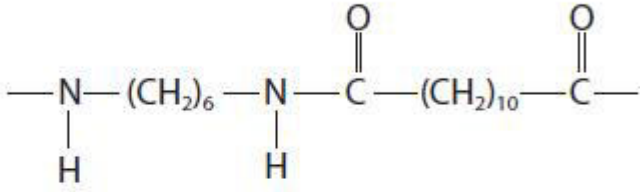
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☐ **A**

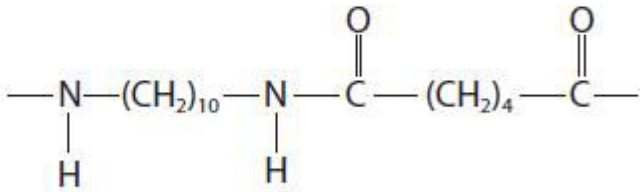


☐ **B**

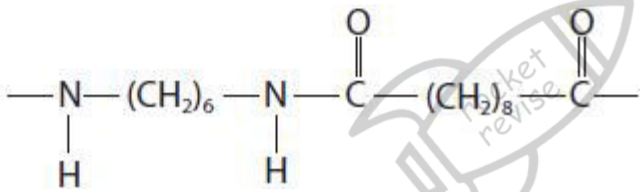


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☐ **C**



☐ **D**



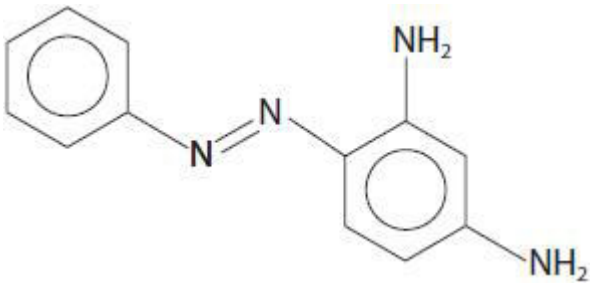
(Total for question = 1 mark)

Q49.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about the azo dye, chrysoidine, which has the structure shown.

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(a) In the production of chrysoidine, phenylamine is first reacted with nitrous acid.

Which equation shows the formation of nitrous acid from sodium nitrite?

(1)

☐ **A** $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$

☐ **B** $NaNO_3 + HCl \rightarrow HNO_3 + NaCl$

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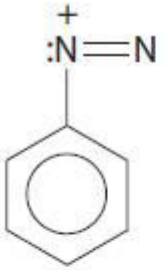
☒ **C** $NaNO_3 + HCl \rightarrow HNO_2 + NaClO$
☒ **D** $NaNO_2 + HCl \rightarrow HNO_2 + Na + ½Cl_2$

(b) The reaction of phenylamine with nitrous acid produces the benzenediazonium ion.

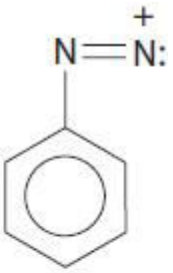
Which is the structure of the benzenediazonium ion?

(1)

☒ **A**

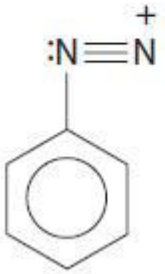


☒ **B**

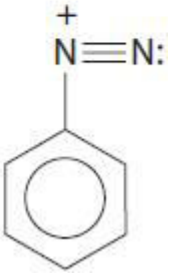


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☒ **C**



☒ **D**



(c) Which reactant is required for the coupling reaction with the benzenediazonium ion to form chrysoidine?

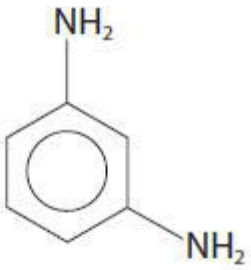
(1)

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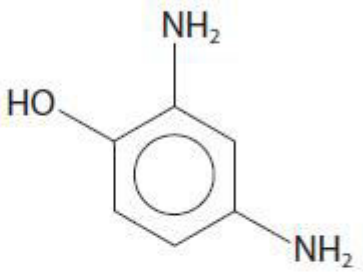
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☐ **A**

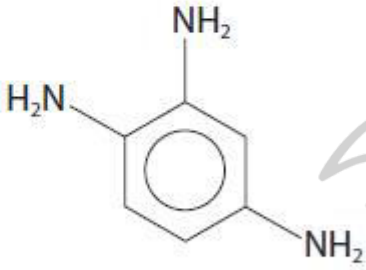


☐ **B**

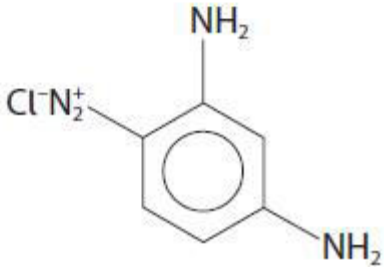


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☐ **C**



☐ **D**



(Total for question = 3 marks)

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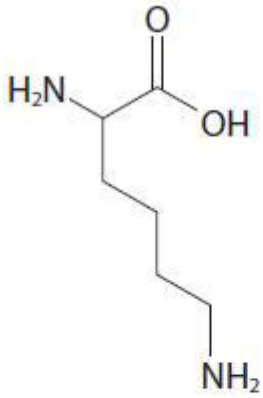
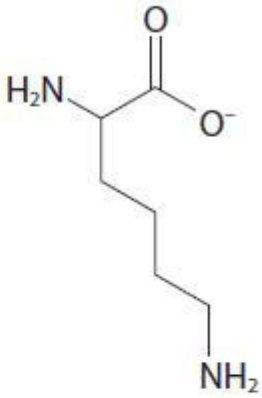
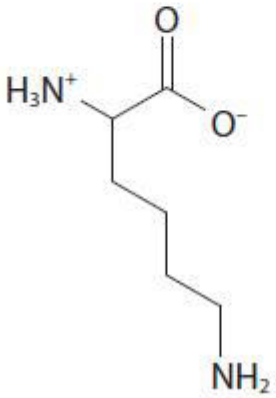
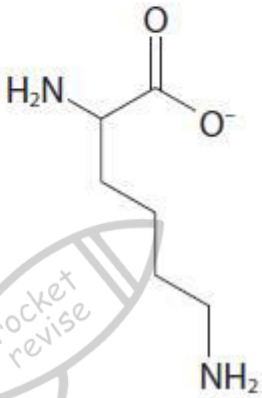
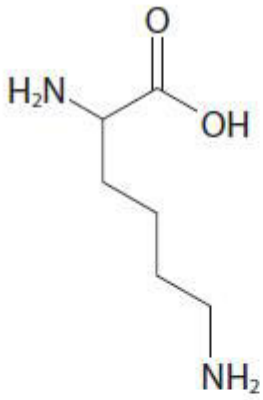
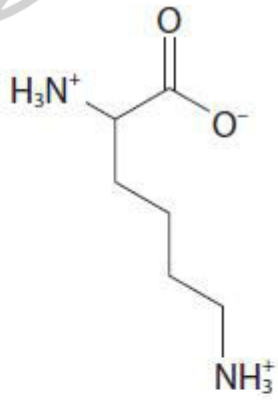
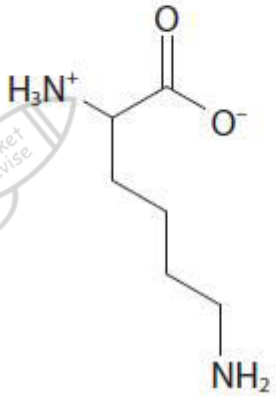
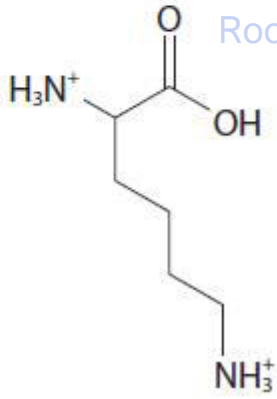
Q50.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which shows the structure of the amino acid lysine as a solid, and in solution at high pH?

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	Solid	High pH solution
☐ A		
☐ B		
☐ C		
☐ D		

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(Total for question = 1 mark)

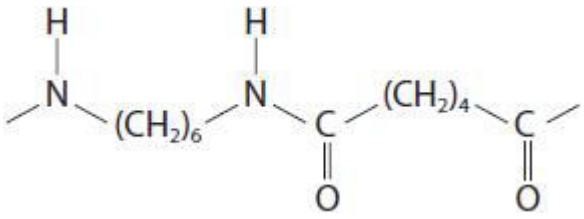
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Q51.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A repeat unit for a polyamide is shown.



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Which pair of monomers makes this polymer?

- ☐ **A** $H_2N(CH_2)_6NH_2$ and $HOOC(CH_2)_4COOH$
- ☐ **B** $NC(CH_2)_6CN$ and $HOOC(CH_2)_4COOH$
- ☐ **C** $H_2N(CH_2)_4NH_2$ and $HOOC(CH_2)_6COOH$
- ☐ **D** $H_2N(CH_2)_6NH_2$ and $HO(CH_2)_6OH$

(Total for question = 1 mark)

Q52.

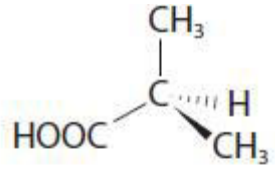
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

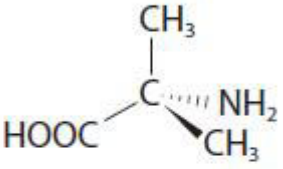
Four compounds (W, X, Y and Z) are shown.

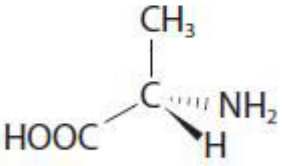
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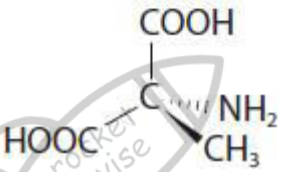
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W 

X 

Y 

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Z 

(a) Which compound is an optically active amino acid?

(1)

- ☒ **A** compound W
- ☒ **B** compound X
- ☒ **C** compound Y
- ☒ **D** compound Z

(b) Which compound does **not** have the same number of peaks in the low-resolution proton NMR spectrum **and** in the ^{13}C NMR spectrum?

(1)

- ☒ **A** compound W
- ☒ **B** compound X
- ☒ **C** compound Y
- ☒ **D** compound Z

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(Total for question = 2 marks)

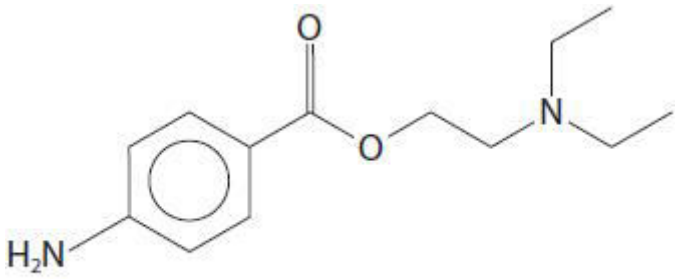
Q53.

Local anaesthetics in dentistry

Procaine is a local anaesthetic that was commonly used until the end of the last century in dental procedures to numb the area around a tooth.

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procaine

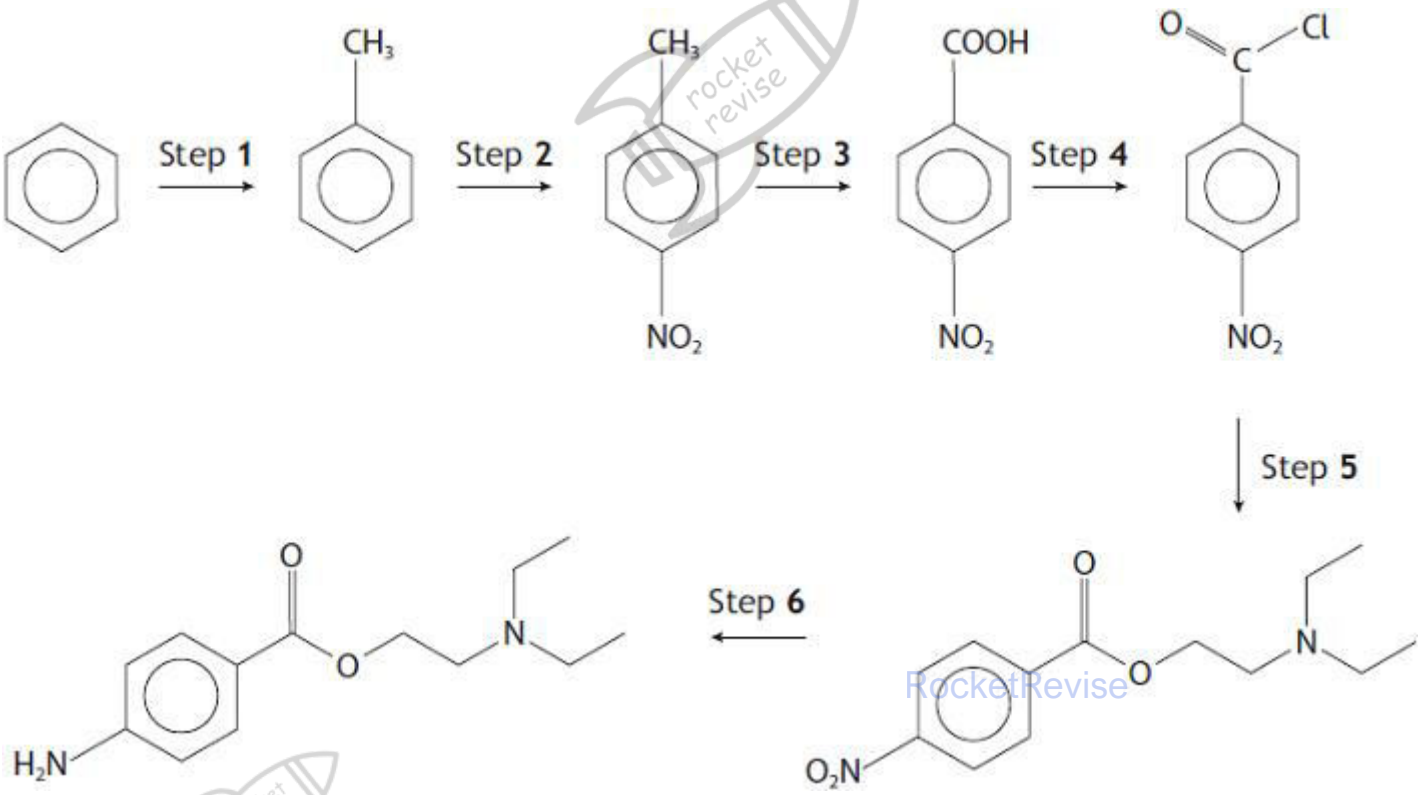
(a) Deduce the molecular formula of procaine.

(1)

.....

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(b) A possible reaction sequence to make procaine is shown.



(i) Identify, by name or formula, the two reagents needed for Step **1**.

(2)

.....

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(ii) For Step **2**, draw the mechanism for this reaction, including the formation of a suitable electrophile and the regeneration of the catalyst. Include curly arrows.

(5)

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(iii) For Step **5**, name the type of reaction taking place and draw the skeletal formula of the reagent needed to carry out this reaction.

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(2)

Type of reaction

.....

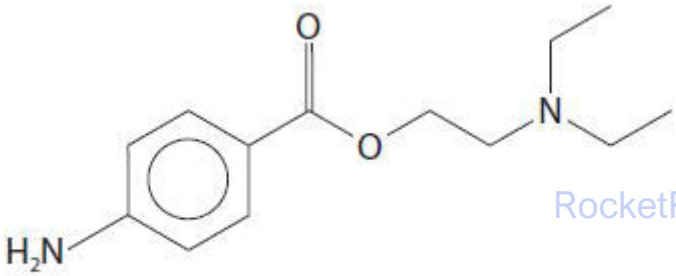
Skeletal formula of the reagent

(iv) Identify, by name or formula, the reagents required for Step **6**.

(1)

.....

(c) Procaine contains two nitrogen atoms, one of which is a stronger base than the other.



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procaine

(i) Explain why the basicity of the nitrogen atoms in procaine is different.

(3)

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(ii) Procaine hydrochloride is made by reacting procaine with hydrogen chloride in a 1 : 1 ratio.
Draw the structure of procaine hydrochloride, showing how the hydrogen ion attaches to the molecule.

(1)

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(iii) Explain why procaine hydrochloride is more soluble in aqueous solutions such as blood plasma than procaine.

(2)

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(d) Nowadays, procaine has been replaced by lidocaine, a related compound, as a local anaesthetic in dentistry.

Lidocaine is supplied in 2.2 cm^3 cartridges that contain 44 mg of lidocaine. A typical adult dose is 1.5 cm^3.
Calculate the number of molecules of lidocaine present in a 1.5 cm^3 dose.
[Molar mass lidocaine = 234 g mol^{-1}]

(3)

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(Total for question = 20 marks)

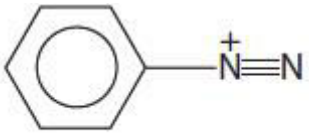
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Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) Which reagents are mixed with hydrochloric acid to produce the benzenediazonium ion?



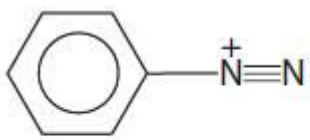
benzenediazonium ion

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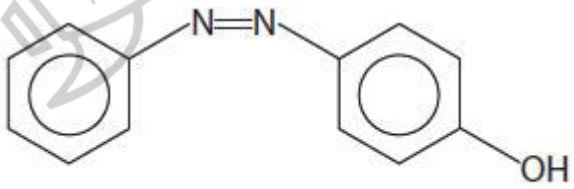
(1)

- ☒ **A** $C_6H_5NH_2$ and $NaNO_2$
- ☒ **B** $C_6H_5NO_2$ and $NaNO_2$
- ☒ **C** $C_6H_5NO_2$ and $NaNO_3$
- ☒ **D** $C_6H_5NH_2$ and $NaNO_3$

(b) Which reagents and conditions are used to prepare the yellow dye 4-phenyldiazenylphenol from the benzenediazonium ion?



benzenediazonium ion



4-phenyldiazenylphenol

(1)

- ☒ **A** phenol dissolved in an acid solution at 25 °C
- ☒ **B** phenol dissolved in an alkaline solution at 25 °C
- ☒ **C** phenol dissolved in an acid solution at 5 °C
- ☒ **D** phenol dissolved in an alkaline solution at 5 °C

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(Total for question = 2 marks)

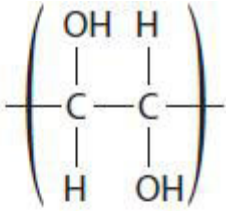
Q55.

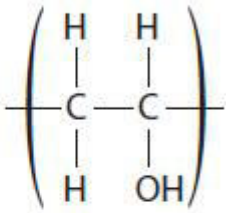
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the repeat unit of the water-soluble polymer poly(ethenol)?

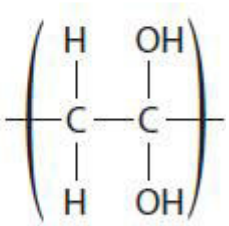
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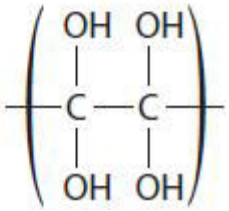
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☒ **A** 

☒ **B** 

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☒ **C** 

☒ **D** 

(Total for question = 1 mark)

Q56.

Trimethylamine, C_3H_9N, is the main reason rotting fish smells so bad. RocketRevise
Trimethylamine is present in the blood and other fluids of many animals but it has higher concentrations in fish.

(a) (i) Draw the **skeletal** structure of trimethylamine.

(1)

(ii) Identify the class of amine to which trimethylamine belongs.

(1)

.....

(iii) Trimethylamine is a gas at room temperature but readily dissolves in water.
Draw a diagram of the intermolecular force that leads to trimethylamine being soluble in water.

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Include relevant dipoles and lone pairs.

(2)

(iv) An aqueous solution of trimethylamine is alkaline.
Write an **ionic** equation to show how trimethylamine acts as a base.
State symbols are not required.

(1)

(b) When trimethylamine oxidises, it forms trimethylamine *N*-oxide, C_3H_9NO.

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Healthy humans have a concentration of 3.90×10^{-6} mol dm^{-3} trimethylamine *N*-oxide in their blood.
An average adult human has 5.00 dm^3 of blood.
Calculate the mass, in grams, of trimethylamine *N*-oxide in an average human's blood, giving your answer to a suitable number of significant figures.

(2)

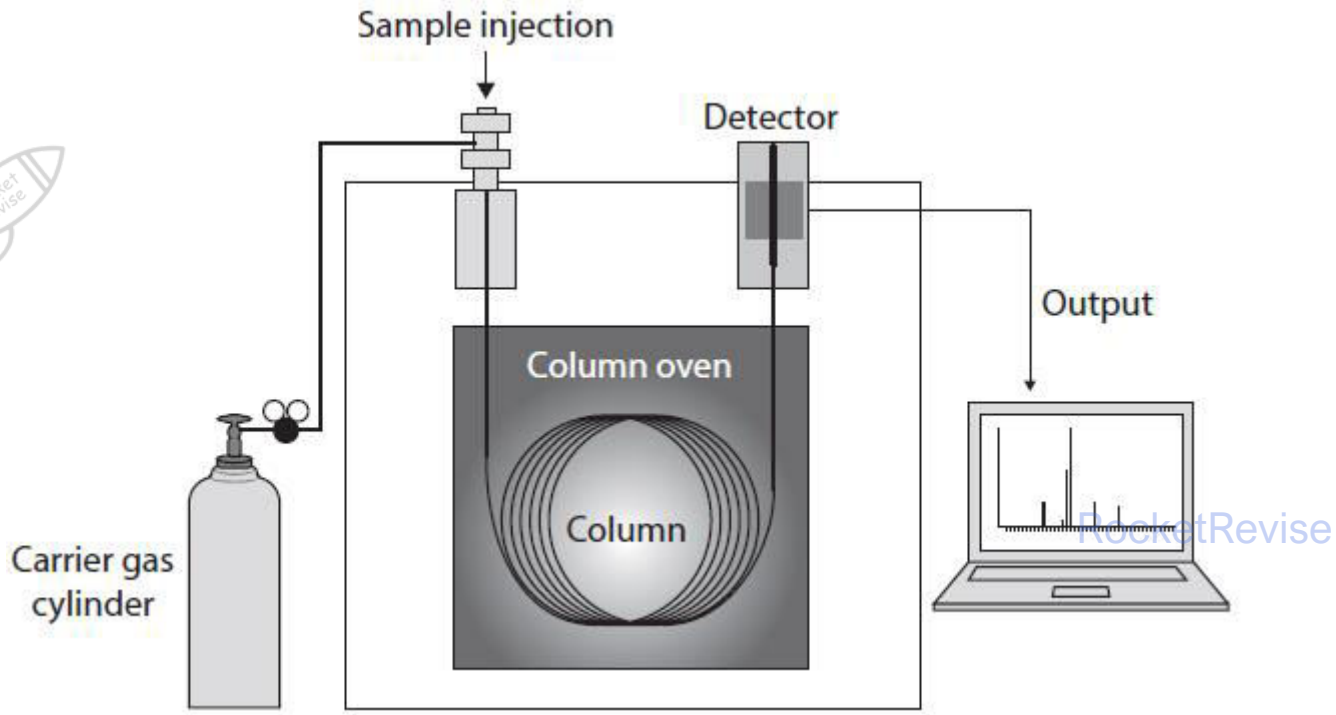
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(c) Trimethylamine is a major air pollutant originating from food waste, farm animals and vehicle exhausts. Trimethylamine levels need to be monitored as it can easily be absorbed through human skin.

Gas chromatography (GC) can be used to measure the level of trimethylamine in air, using the apparatus shown in the diagram.

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(i) Briefly describe how trimethylamine will be **separated** from other chemicals present in the sample of air.

(2)

.....

.....

.....

.....

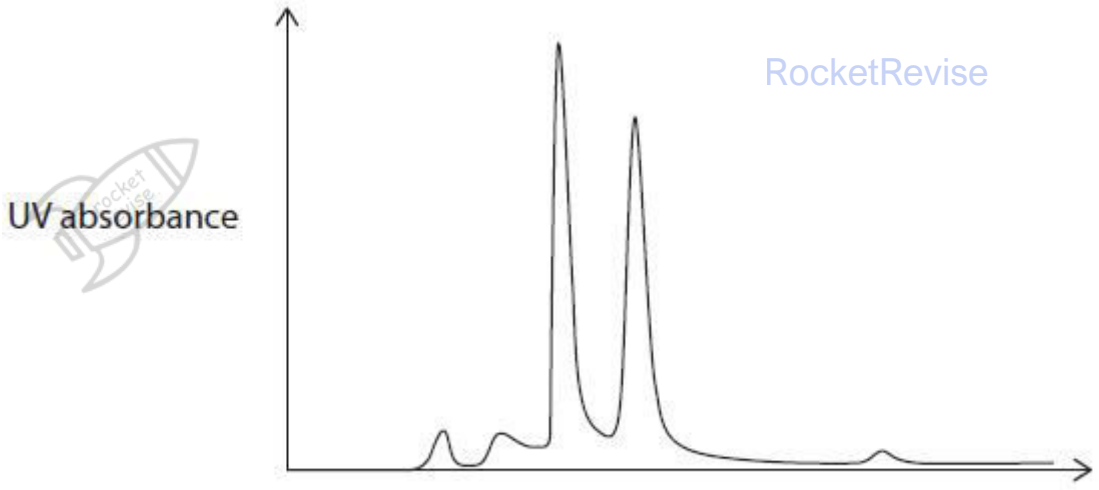
.....

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(ii) High-performance liquid chromatography (HPLC) is used to measure levels of trimethylamine and trimethylamine *N*-oxide in blood and urine.

Label the x-axis on the chromatogram.

(1)



.....

(iii) Name the analytical technique that can be used with either GC or HPLC to identify the compound

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responsible for each peak.

(1)

.....

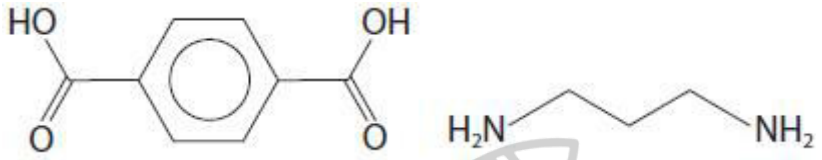
(Total for question = 11 marks)

Q57.

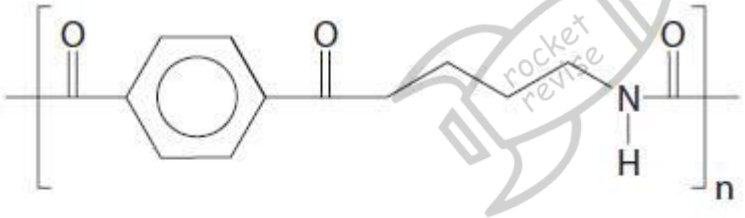
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

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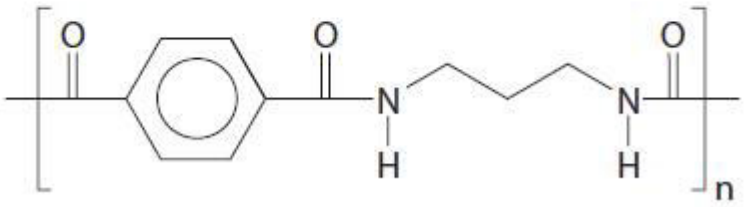
Which is the repeat unit of the polymer formed from the monomers shown?



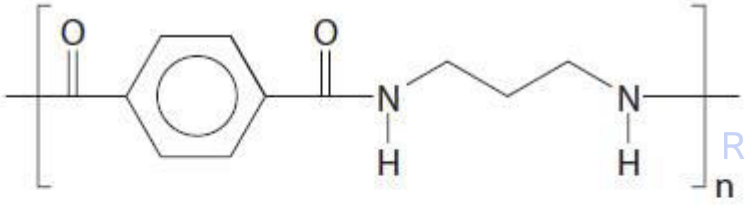
☐ **A**



☐ **B**

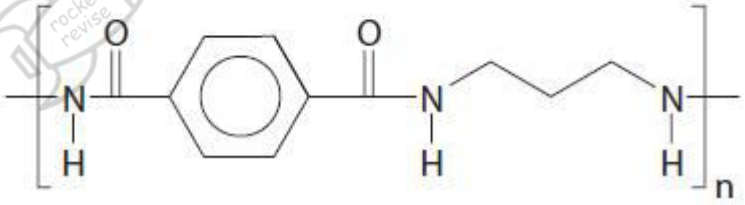


☐ **C**



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☐ **D**



(Total for question = 1 mark)

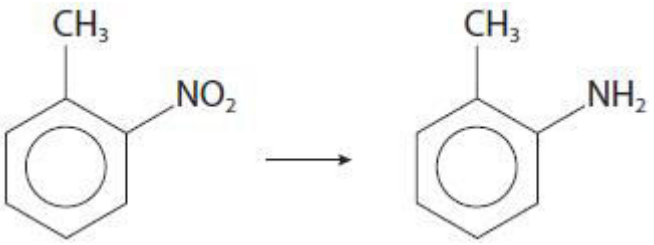
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Q58.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

Which are the reagents and conditions for the reaction shown?



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- ☐ **A** $K_2Cr_2O_7$, H^+ and reflux
- ☐ **B** $LiAlH_4$ in dry ether
- ☐ **C** H_2 gas at high temperature
- ☐ **D** Sn, concentrated HCl and reflux

(Total for question = 1 mark)

Q59.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

Which product forms when butylamine reacts with ethanoyl chloride?

- ☐ **A** ethanamide
- ☐ **B** butanamide
- ☐ **C** *N*-butylethanamide
- ☐ **D** *N*-ethylbutanamide

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(Total for question = 1 mark)

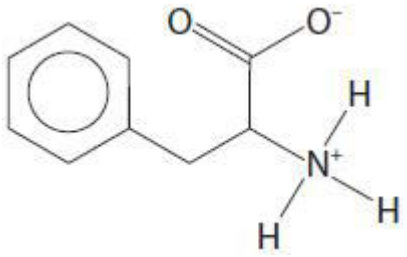
Q60.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

At what pH would the amino acid phenylalanine have the structure shown?

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- ☐ **A** pH 2
- ☐ **B** pH 6
- ☐ **C** pH 10
- ☐ **D** pH 14

(Total for question = 1 mark)

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